

SingleRAN

FMA Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Added descriptions of NE analysis with the KPI period being **P60** or other values to [3.1.2.2.4 All Scenario KPI Overview](#) and [3.1.2.3.4 All Scenario KPI Overview](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-07-30).

Technical Changes

Change Description	Parameter Change
Optimized the KPI measurement and display on the All Scenario KPI Overview tab page. For details, see 3.1.2.2.4 All Scenario KPI Overview and 3.1.2.3.4 All Scenario KPI Overview .	None

Change Description	Parameter Change
Added support for displaying automatic detection results collected by the NIC on the FMA's Information Collection tab page. For details, see 3.1.2.2.3 Information Collection and 3.1.2.3.3 Information Collection .	None
Added incompatibility of BTS5929R with the FMA. For details, see 3.3.3 Hardware .	None
Added RedCap-, access-, and data transmission-related counters for network-wide automatic detection in all scenarios. For details, see 3.1.2.3.7 Other Functions .	None
Deleted the auxiliary fault recovery function from LTE wireless fault management.	None
Added the functions of analyzing and exporting power information to LTE fast fault diagnosis. For details, see 3.1.2.2.2 Fast Fault Diagnosis .	None
Added support for the MOCN scenario by all scenario KPI overview. For details, see: • 3.1.2.2.4 All Scenario KPI Overview • 3.1.2.3.4 All Scenario KPI Overview	None
Added support for analysis of RedCap-related performance counters by all scenario KPI overview in NR. For details, see 3.1.2.3.4 All Scenario KPI Overview .	None
Added detailed causes to OMCH Maintenance Info for the transmission FMA. For details, see 3.1.3.2 Fast Fault Diagnosis .	None

Editorial Changes

Changed **NR cell ID** to **NR DU cell ID** in the key cell information table, which is part of the diagnosis information in interference fault scenarios. For details, see [3.1.2.3.2 Fast Fault Diagnosis](#).

Deleted the statement that the cell dashboard displays cell data of a maximum of five KPIs from [3.1.2.2.2 Fast Fault Diagnosis](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed, such as the field names and figures.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, New Radio (NR), and NB-IoT.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
MMFD-2010 000709	Fault Management Assistance (FMA)	3.1 Principles
		3.4 Operation and Maintenance

3 FMA

3.1 Principles

3.1.1 Overview

3.1.1.1 Background

On a radio access network (RAN), there are many complicated network elements (NEs). This leads to the difficulty in fault locating and low efficiency in fault recovery.

3.1.1.2 Introduction

Fault Management Assistance (FMA) allows users to quickly analyze fault causes and perform troubleshooting when network faults or major emergencies occur.

Table 3-1 describes the functions that the FMA supports. These functions can be used together to locate wireless faults and transmission faults.

Table 3-1 FMA functions

Function	Description	Wireless FM (GSM & UMTS)	Wireless FM (LTE)	Wireless FM (NR)	Trans FM
Fault overview	Provides a visualized fault information overview, including the system operating status. Once a fault occurs, users can quickly obtain system fault information and engage in troubleshooting procedures.	√	√	√	√
Fast fault diagnosis	Provides a comprehensive analysis of a persistent fault on the network based on diagnosis rules, including configurations, performance counters, alarms, and operation logs. An analysis report is generated to help users quickly rectify the network fault.	√	√	√	√

Function	Description	Wireless FM (GSM & UMTS)	Wireless FM (LTE)	Wireless FM (NR)	Trans FM
Hierarchical delimitation	First analyzes abnormal network KPIs and breaks down the fault object based on the protocol layers to identify and define the issue into the minimum object. The performance counters, alarms, status, and operation logs of the identifiable object are then displayed, faults are identified, and a fault analysis report is offered to users.	√	x	x	x

Function	Description	Wireless FM (GSM & UMTS)	Wireless FM (LTE)	Wireless FM (NR)	Trans FM
Information collection	Collects onsite information for fault troubleshooting. Due to fault information enormity, fault information collection is complicated and easily leads to incorrect results and repeated collections, which prolongs the fault recovery time. This function can quickly and accurately collect onsite information, which helps to reduce the time required for fault recovery. NOTE Fault information collected using the FMA is anonymized to protect user privacy.	√	√	√	√
Auxiliary fault recovery	Provides fault recovery assistance for specific configuration faults displayed in the fast fault diagnosis results.	x	x	x	√

Function	Description	Wireless FM (GSM & UMTS)	Wireless FM (LTE)	Wireless FM (NR)	Trans FM
Fault recovery confirmation	Allows users to observe KPIs of NEs in real time and checks whether the fault has been rectified based on KPI observations after a fault recovery solution is implemented.	√	√	√	√
All scenario KPI overview	Provides visualized display of KPI comparison and analysis and allows users to perform site-level KPI and performance counter analysis of NEs managed by the MAE.	x	√	√	x
Service query	Checks whether the service has been restored after a fault recovery solution is implemented. This function only applies to GSM and UMTS networks.	√	x	x	x
Service verification	Rapidly verifies the network status after a network reconstruction is implemented on the live network. This function only applies to GSM and UMTS networks.	√	x	x	x

Function	Description	Wireless FM (GSM & UMTS)	Wireless FM (LTE)	Wireless FM (NR)	Trans FM
NOTE X: not supported; √: supported					

 NOTE

For base stations in integrated deployment mode, different content is displayed for wireless faults and transmission faults when the functions are supported by both wireless and transmission fault management (such as fault overview).
The BSC6960 does not support wireless fault management.

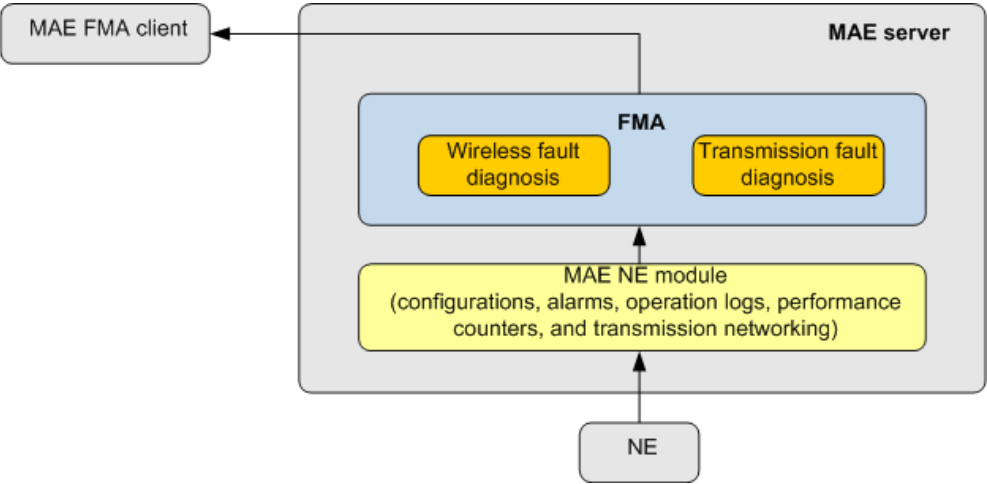
3.1.1.3 Architecture

FMA is deployed on different platforms according to radio access technologies (RATs) and fault types.
FMA functions are deployed as follows:

- For GSM and UMTS, wireless faults are analyzed on the BSC LMT and transmission faults are analyzed on the MAE.
- For LTE and NR, wireless faults and transmission faults are both analyzed on the MAE.

Figure 3-1 shows the FMA architecture on the MAE, which is similar to that on the LMT.

Figure 3-1 FMA architecture on the MAE



3.1.2 Wireless Fault Management

3.1.2.1 GSM and UMTS Wireless Fault Management

GSM and UMTS wireless fault management aims to rectify network faults and verify services. It includes the following functions:

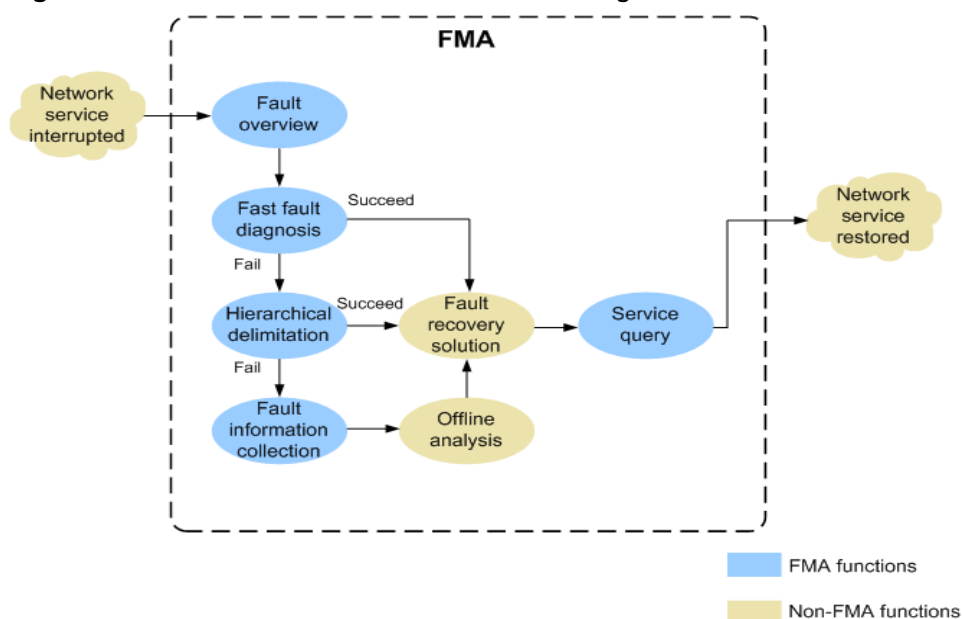
- [3.1.3.1 Fault Overview](#)
- [3.1.2.1.2 Fast Fault Diagnosis](#)
- [3.1.2.1.3 Hierarchical Delimitation](#)
- [3.1.2.1.4 Information Collection](#)
- [3.1.2.1.5 Automatic Fault Diagnosis](#)
- [3.1.2.1.6 Recovery Confirmation](#)
- [3.1.2.1.7 Service Query](#)
- [3.1.2.1.8 Service Verification](#)

 NOTE

- Of these functions, fast fault diagnosis and hierarchical delimitation are applicable to faults that are generated or cleared within the last 8 hours.
- Fast fault diagnosis and hierarchical delimitation cannot be performed at the same time.
- While information collection is in process, do not run the **COL LOG** command to collect logs.

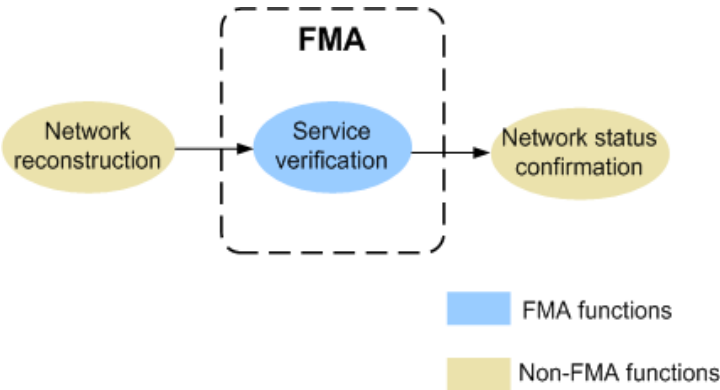
[Figure 3-2](#) shows the troubleshooting process.

Figure 3-2 GSM and UMTS wireless fault management



[Figure 3-3](#) shows the service verification process of GSM and UMTS FMA after a network reconstruction.

Figure 3-3 Service verification process after a network reconstruction



GSM and UMTS FMA is implemented on the LMT. This section describes the principles and application scenarios of GSM and UMTS FMA. For details on the operation procedure, see *BSC6900 GU LMT User Guide* and *BSC6910 GU LMT User Guide*.

NOTE

- Ensure that only one LMT performs the FMA function at a time.
- The GSM and UMTS wireless fault management is not applicable to eGBTSs.

3.1.2.1.1 Fault Overview

Fault overview provides visualized illustration of operating status and fault information about GSM and UMTS networks. This function allows users to quickly identify a fault and start fault troubleshooting. [Table 3-2](#) describes the information that users can obtain using this function.

Table 3-2 Fault overview user information

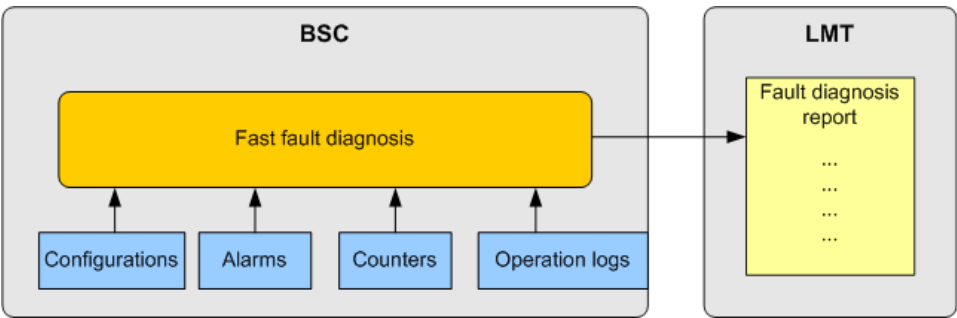
Category	Description
UMTS KPI	Trends of the following KPIs: RRC Connection Setup, CS RAB Setup, PS RAB Setup, CS Erlang, PS Traffic, CS Service Drop, PS Service Drop, Paging, and SCCP Connect Setup Success Rate
GSM KPI	Trends of the following KPIs: CS Call Setup Success Rate, CS Immediate Assignment Success Rate, CS Assignment Success Rate, Uplink PS TBF Establishment Success Rate, Downlink PS TBF Establishment Success Rate, CS TCH Call Drop Rate, Uplink PS TBF Drop Rate, Downlink PS TBF Drop Rate, CS Erlang, Uplink PS Traffic, Downlink PS Traffic, Average Call Duration, Channel Requests, Assignment Requests, Location Updating Channel Requests, and SCCP Connect Setup Success Rate

Category	Description
KPI trend chart	A KPI trend chart that contains three curves illustrating KPI changes in the last eight hours, the same hours from the previous day, and during the same hours on the same day from the previous week, respectively. For example, assume that it is now midnight (24:00) on Tuesday. In this case, the three curves show the KPI changes during the following periods, respectively: 16:00 to 24:00 today, 16:00 to 24:00 yesterday, and 16:00 to 24:00 last Tuesday. NOTE The horizontal X-coordinate represents the measurement time (expressed in the format of HH:MM), and the vertical Y-coordinate represents the counter value, in corresponding respective units. When fast fault diagnosis is used to analyze performance data, performance data generated during a sampling period is not available if the measurement state was DISABLED in the sampling period.
KPI trend value	- Automatically displays the KPI value corresponding to a point on the KPI change trend curve when users select a point using the mouse pointer. - Provides KPI filtering on the right of the fault overview interface to allow viewing the change trend of the selected KPI.

3.1.2.1.2 Fast Fault Diagnosis

This fast fault diagnosis function comprehensively analyzes network faults from the perspectives of counters, alarms, configurations, and operation logs based on the diagnosis rules and then generates a fault diagnosis report. [Figure 3-4](#) illustrates the principle of fast fault diagnosis.

Figure 3-4 Fast fault diagnosis



For details on fast fault diagnosis, see *BSC6900 GU LMT User Guide* and *BSC6910 GU LMT User Guide*.

Wireless Fault Diagnosis Scenarios

[Table 3-3](#) lists the scenarios of GSM and UMTS wireless fault diagnosis.

Table 3-3 Wireless fault diagnosis scenarios

Fault Diagnosis	Scenario Option	
UMTS fault diagnosis	KPI	RRC Success Rate
		CS RAB Setup Success Rate
		CS Call Drop Rate
		PS RAB Setup Success Rate
		PS Service Drop Rate
		Paging
	Traffic	CS Erlang
		PS Traffic
	Transmission	Site Break Down
		Transmission Signaling Analysis
		Transmission Connectivity Analysis
	Board	Board Faults Diagnosis
		Communication Failure Between NE and OSS
		OMU Faults Diagnosis
		Analysis on the Proportion of Faulty Board Links
		Hardware Sub-health Diagnosis
	Health check	Equipment Health Check
	Others	License Faults Diagnosis
		RNC in Pool Load Sharing
GSM fault diagnosis	KPI	GSM CS Immediate Assignment Success Rate
		GSM CS Assignment Success Rate
		GSM CS Call Setup Success Rate
		GSM TCH Call Drop Rate
		GSM PS TBF Establish Success Rate
		GSM PS TBF Drop Rate
		GSM PS TBF Drop Rate
		GSM Paging Success Rate

Fault Diagnosis	Scenario Option	
		GSM Location Update Success Rate
		GSM Outgoing Inter-Cell Handover Decrease
	Transmission	GSM BTS Interruption
		GSM Ater Interface Interruption
		GSM A Interface Interruption
		GSM GB Interface Interruption
		Transmission Connectivity Analysis
	Traffic	GSM CS Traffic
		GSM GB Interface Traffic
	Board	GSM Repeated BSC Board Faults
		Board Faults Diagnosis
		Communication Failure Between NE and OSS
		OMU Faults Diagnosis
		Hardware Sub-health Diagnosis
	Health check	GSM CS Key Parameters Check
		GSM PS Key Parameters Check
		GSM BSC Board Configuration Check
		GSM Equipment Health Check
	Others	GSM PDCH Fault
		GSM Voice Fault Analyse

Table 3-4, Table 3-5, and Table 3-6 list the fault diagnosis scenarios for RNC in Pool.

Table 3-4 Fault diagnosis scenarios for RNC in Pool (supporting RNC in Pool Load Sharing only)

Diagnosis Item	Load Sharing Type = MASTER	Load Sharing Type = OVERFLOW
RRC connection setup	√	x
CS service setup	√	x

Diagnosis Item	Load Sharing Type = MASTER	Load Sharing Type = OVERFLOW
CS call drop	√	x
PS service setup	√	x
PS call drop	√	x
CS Erlang	√	x
PS traffic	√	x
Paging	√	x
A large number of unavailable cells	√	x
Equipment health check	√	√
RNC in Pool load sharing	√	x
NOTE X: not supported; √: supported		

Table 3-5 Fault diagnosis scenarios for RNC in Pool (supporting RNC in Pool Redundancy only)

Diagnosis Item	Host Status = MASTER	Host Status = BACKUP
RRC connection setup	√	x
CS service setup	√	x
CS call drop	√	x
PS service setup	√	x
PS call drop	√	x
CS Erlang	√	x
PS traffic	√	x
Paging	√	x
A large number of unavailable cells	√	x
Equipment health check	√	√
RNC in Pool load sharing	x	x
NOTE X: not supported; √: supported		

Table 3-6 Fault diagnosis scenarios for RNC in Pool (supporting RNC in Pool Load Sharing and Redundancy)

Diagnosis Item	Load Sharing Type = MASTER and Host Status = MASTER	Load Sharing Type = OVERFLOW and Host Status = BACKUP	Load Sharing Type = OVERFLOW and Host Status = MASTER
RRC connection setup	√	x	x
CS service setup	√	x	x
CS call drop	√	x	x
PS service setup	√	x	x
PS call drop	√	x	x
CS Erlang	√	x	x
PS traffic	√	x	x
Paging	√	x	x
A large number of unavailable cells	√	x	x
Equipment health check	√	√	√
RNC in Pool load sharing	√	x	x
NOTE X: not supported; √: supported			

Fault Diagnosis Report

A fast fault diagnosis report provides the following information: RNC basic information, KPI trend, FMA dashboard, fault analysis report, and operation logs of the last 24 hours. For details, see *BSC6900 GU LMT User Guide* and *BSC6910 GU LMT User Guide*.

3.1.2.1.3 Hierarchical Delimitation

Hierarchical delimitation can be used to analyze the fault that cannot be identified by the fast fault diagnosis function.

Fault delimitation analyzes abnormal network KPIs (as listed in [Table 3-2](#)) based on standard protocol layers. This is to locate and identify minimum objects

attributable to the network fault, identify and display counters, alarms, status, and operation logs related to these objects, and finally generate a fault analysis report. You can query historical analysis reports by selecting the storage duration.

Table 3-7 and **Table 3-8** list abnormal UMTS and GSM network KPIs, respectively.

Table 3-7 Abnormal UMTS network KPIs

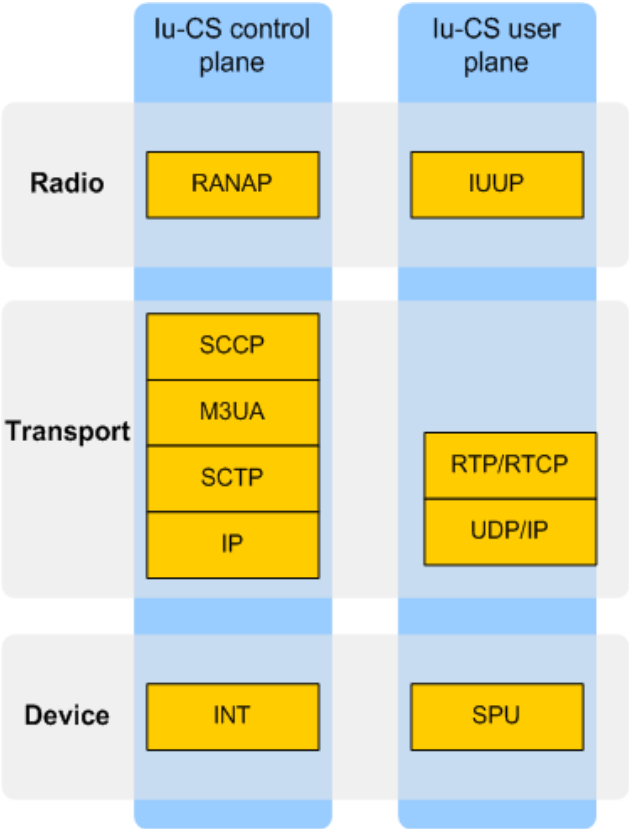
KPI	Interface Type	Control Plane or User Plane
RRC Connection Setup	Iub	Control plane
CS RAB Setup	Iub/Iu-CS	Control plane/User plane
PS RAB Setup	Iub/Iu-PS	Control plane/User plane
CS Traffic	Iub/Iu-CS	Control plane/User plane
PS Traffic	Iub/Iu-PS	Control plane/User plane
CS Service Drop	Iub/Iu-CS	Control plane/User plane
PS Service Drop	Iub/Iu-PS	Control plane/User plane
Paging	Iub/Iu	Control plane

Table 3-8 Abnormal GSM network KPIs

Fault Indicator	Interface Type	Control Plane/User Plane
CS Call Setup Success Rate	Abis/A	Control plane/User plane
CS Immediate Assignment Success Rate	Abis	Control plane
CS Assignment Success Rate	Abis/A	Control plane/User plane
TBF Establishment Success Rate	Abis/Gb	Control plane/User plane
TCH Call Drop Rate	Abis/A	Control plane/User plane
TBF Drop Rate	Abis/Gb	Control plane/User plane
CS Erlang	Abis/A	Control plane/User plane
PS Traffic	Abis/Gb	Control plane/User plane

Figure 3-5 shows an example of the identification process of the Iu-CS interface on an IP-based UMTS network based on the protocol stack.

Figure 3-5 IP protocol stack for the lu-CS interface



The information contained in a hierarchical delimitation report is described in the following table.

Table 3-9 Hierarchical delimitation report information

Item	Description
Faulty cells	Information about cells affected by a fault is provided. The following information is included: • Cell ID: ID of the faulty cell • Failure rate: proportion of the number of failures in a faulty cell to that in a base station controller (BSC) related to a specific counter • Fault counter: proportion of the number of failures in a faulty cell to the number of attempts in this cell related to a specific counter • Number of attempts: number of attempts in this cell related to a specific counter • Base station name: name of the base station to which the faulty cell belongs • Interface board: interface board where the faulty cell is configured • Subsystem: a subsystem where the faulty cell is configured

Item	Description
Scenario selection	Provides the minimum objects after faults are analyzed based on standard protocol layers. Users can select a fault object to view related KPIs, alarms, and status. The fault objects are displayed below. The interface between BSC and BTS (Iub/Abis interface) • Control plane – Wireless layer objects (NBAP) – Transmission layer objects (SCTPLNK and ETHPORT) – Device layer objects (INT board and control plane subsystem) • User plane – Wireless layer objects (Abis) – Transmission layer objects (ETHPORT) – Device layer objects (INT board) The interface between BSC and CN (Iu/A/Gb interface) • Control plane – Wireless layer objects (RANAP) – Transmission layer objects (MTP3LINK, SCTPLNK, and ETHPORT) – Device layer objects (INT board and control plane subsystem) • User plane – Wireless layer objects (IUUP and GTPU) – Transmission layer objects (ETHPORT) – Device layer objects (INT board)

3.1.2.1.4 Information Collection

If a fault cannot be located using the fast fault diagnosis and hierarchical delimitation functions, onsite fault information must be obtained for fault troubleshooting. Two methods can be used to collect fault information: fast information collection and user-defined information collection.

Fast Information Collection

The information collected using fast information collection varies with the NE, as described in [Table 3-10](#).

Table 3-10 Fast information collection

NE	Collection Batch	Data Collection File	Data Collection Range
GSM BSC	First batch	Counter result files	4 hours
		Historical alarm files	3 days
		Operation logs	48 hours
		BSC data configuration files and MML command outputs	Full data
		BSC board last word logs	Within 20 minutes if last word logs are generated
	Second batch	Common debugging logs	Between the start time and end time selected by users
		GSM CS single-user CHR logs	Between the start time and end time selected by users
		GSM CHR logs	Between the start time and end time selected by users
		GSM one-way audio logs	Between the start time and end time selected by users
		Host running logs	Between the start time and end time selected by users
		MML command outputs	Full data
		Last word log files	One day
		Platform CHR logs	Between the start time and end time selected by users
		Transmission path logs	1 hour
		GBTS deployment logs	1 hour
RNC	First batch	Historical alarm files	48 hours
		Operation logs	48 hours
		BSC data configuration files	Full data
		UKPI snapshots	5 minutes or 24 hours

NE	Collection Batch	Data Collection File	Data Collection Range
	Second batch	BSC board last word logs	Within 20 minutes if last word logs are generated
		Counter result files	2 hours
		UMTS CHR logs	One file for each subrack
		Common debugging logs	One file for each subrack
		CALLFAULT logs	One file for each subrack

User-defined Information Collection

User-defined information collection allows user to customize fault information items before information collection. This information collection mode facilitates routine O&M.

The following items can be defined by users: the start time and end time of data collection, collected log file types, and MML query commands.

3.1.2.1.5 Automatic Fault Diagnosis

Automatic Startup on FMA

The FMA periodically detects KPIs and fault alarms and automatically starts fault diagnosis based on the preset thresholds. [Table 3-11](#), [Table 3-12](#), [Table 3-13](#), and [Table 3-14](#) list the triggering conditions.

Table 3-11 Automatic startup on FMA triggered by KPI thresholds (UMTS)

Scenario	KPI Name	Default Threshold
UMTS fault diagnosis	RRC Setup Success Ratio	95%
	CS RAB Setup Success Ratio	95%
	PS RAB Setup Success Ratio	95%
	HSDPA Setup Success Ratio	95%
	HSUPA Setup Success Ratio	95%
	CS Call Drop Ratio	2%

Scenario	KPI Name	Default Threshold
	PS Call Drop Ratio	5%
	Security Mode Success Ratio	95%
	Location Update Success Ratio	80%
	Route Area Success Ratio	80%
	Attach Success Ratio	70%
	PDP Activation Success Ratio	70%
	CS Authentication Success Ratio	90%
	PS Authentication Success Ratio	90%
	CS Service Request Success Ratio	90%
	PS Service Request Success Ratio	90%

Table 3-12 Automatic startup on FMA triggered by KPI thresholds (GSM)

Scenario	KPI Name	Default Threshold
GSM fault diagnosis	Assignment Success Rate	90%
	Call Establishment Success Rate	80%
	Location Update Success Rate	80%
	Call Drop Rate on TCH per Cell	5%
	Uplink TBF Establishment Success Rate	65%
	Downlink TBF Establishment Success Rate	65%
	Uplink EGPRS TBF Establishment Success Rate	65%

Scenario	KPI Name	Default Threshold
	Downlink EGPRS TBF Establishment Success Rate	65%
	Uplink TBF Drop Rate	10%
	Downlink TBF Drop Rate	10%
	Uplink EGPRS TBF Drop Rate	10%
	Downlink EGPRS TBF Drop Rate	10%

Table 3-13 Automatic startup on FMA triggered by alarms (UMTS)

Scenario	Product Name	Alarm Name
UMTS fault diagnosis	BSC6900/BSC6910	ALM-21531 SAAL Link Fault
		ALM-21522 Low SCCP Setup Success Rate
		ALM-21521 SCCP Subsystem Prohibited
		ALM-21552 M3UA Destination Entity Route Unavailable
		ALM-21503 MTP3 DSP Inaccessible
		ALM-21541 SCTP Link Fault
		ALM-20241 Board Unavailable
	BSC6900	ALM-20737 OMU Disconnected from Host

Table 3-14 Automatic startup on FMA triggered by alarms (GSM)

Scenario	Product Name	Alarm Name
GSM fault diagnosis	BSC6900/BSC6910	ALM-20804 BTS Unavailable
	BSC6900/BSC6910	ALM-21807 OML Fault
	BSC6900/BSC6910	ALM-21825 CSL Link Fault
	BSC6900/BSC6910	ALM-22009 NSVL Dynamic Configuration Procedure Failure

Scenario	Product Name	Alarm Name
	BSC6900/BSC6910	ALM-22005 NSE Faulty
	BSC6900/BSC6910	ALM-22008 PTP BVC Faulty
	BSC6900/BSC6910	ALM-21521 SCCP Subsystem Prohibited
	BSC6900/BSC6910	ALM-21552 M3UA Destination Entity Route Unavailable
	BSC6900/BSC6910	ALM-21503 MTP3 DSP Inaccessible
	BSC6900	ALM-20246 Ater RSL Faulty
	BSC6900	ALM-20261 Ater OML Faulty
	BSC6900/BSC6910	ALM-21541 SCTP Link Fault
	BSC6900/BSC6910	ALM-20241 Board Unavailable
	BSC6900/BSC6910	ALM-20737 OMU Disconnected from Host

Automatic Tracing Startup

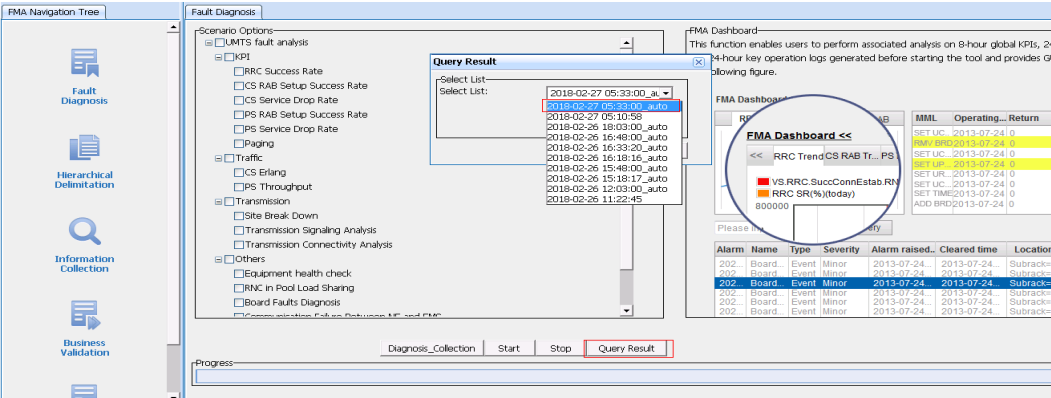
When the following KPIs reach the threshold of automatic startup on FMA, the FMA automatically starts tracing signaling over the lu interface (including SCCP) based on specific fault scenarios to help locate faults over this interface. You can use the **Download Source Data** function to download the automatic tracing file to a local disk.

- The value of CS/PS RAB Setup Success Rate decreases.
- The value of CS/PS Service Drop Rate increases.

Automatic Startup Report Acquisition

1. Log in to the LMT and click the **FMA** button to enter the **FMA** tab page.
2. Click **Query Result** in the lower part of the **FMA** tab page, as shown in [Figure 3-6](#).
3. In the displayed **Query Result** dialog box, select the time range from the drop-down list box. In this list box, the files with a suffix of "auto" are automatic diagnosis reports, and the rest are manual diagnosis reports.
4. Click **Start** and wait for the generation of a fault diagnosis report.
5. Click **Save Report** or **Download Source Data** to download the analysis report and collected data to a local disk.

Figure 3-6 Acquiring the automatic startup report



3.1.2.1.6 Recovery Confirmation

On the fault overview page, users can view the running status of the GSM and UMTS networks after service recovery solutions are implemented to check whether a fault is rectified. For details, see [3.1.2.1.1 Fault Overview](#).

3.1.2.1.7 Service Query

This function quickly checks whether services have been restored after a fault recovery solution is implemented.

Due to the long period of normal KPI checks, the status of restored services is hardly acquired in time. This sub-function supports the service KPI query within a short period (5 minutes), allowing users to rapidly obtain the status of restored services.

3.1.2.1.8 Service Verification

This function rapidly queries the network status after a network reconstruction is implemented on the live network.

This function compares performance indicators, alarms, and configuration before and after a network reconstruction, and displays the service status to rapidly determine whether a service operates properly after the network reconstruction.

[Table 3-15](#) and [Table 3-16](#) list applicable UMTS and GSM network reconstruction scenarios and check items, respectively.

Table 3-15 Applicable UMTS network reconstruction scenarios and check items

Scenario	Check Item
Common indicator verification	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate

Scenario	Check Item
	Traffic volume
Transmission mode transformation from ATM to IP over the lub interface	Transmission link status
	Cell status
	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate
	Activity factor
	Port protection solution
Rehomming	Transmission link status
	Cell status
	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate
	Activity factor
	Port protection solution
lur interface reconstruction	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate
	Traffic volume
	Transmission link status
	Migration success rate
lu interface reconstruction	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate

Scenario	Check Item
	CS call drop rate
	PS call drop rate
	Traffic volume
	Transmission link status
	NAS-layer indicators over the lu interface
RNC license replacement	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate
	Traffic volume
IUFLEX	RRC connection setup success rate
	CS RAB setup success rate
	PS RAB setup success rate
	CS call drop rate
	PS call drop rate
	Traffic volume
	Transmission link status
	Configuration integrity
	NAS-layer indicators over the lu interface

Table 3-16 Applicable GSM network reconstruction scenarios and check items

Scenario	Check Item
IP-based restructuring over the A interface	Control-plane alarms over the A interface
	User-plane alarms over the A interface
	CS access-related indicators
	CS call drop rate
	Channel congestion-related indicators
	IP path-related indicators

Scenario	Check Item
MSC Pool reconstruction	Control-plane alarms over the A interface
	Alarms related to core network flow control and congestion
	CS access-related indicators
	CS call drop rate
	Channel congestion-related indicators
	MSC Pool selection-related indicators
IP-based restructuring over the Gb interface	NSVL link-related alarms
	PS service-related alarms
	TBF establishment success rate
	TBF call drop rate
	PS traffic-related indicators
SGSN Pool reconstruction	NSVL/NSVC link-related alarms
	PS service-related alarms
	TBF establishment success rate
	TBF call drop rate
	PS traffic-related indicators
Base station cutover	Alarms related to disconnection with a base station and out-of-service of a cell
	Invalid voice service alarm
	PS service-related alarms
	CS access-related indicators
	CS call drop rate
	Channel congestion-related indicators
	TBF establishment success rate
	TBF call drop rate
	PS traffic-related indicators

3.1.2.2 LTE Wireless Fault Management

LTE wireless fault management is implemented on the MAE to help users troubleshoot wireless faults on LTE networks.

LTE wireless fault management obtains eNodeB configurations (the size of a configuration file does not exceed 200 MB), alarms, performance counters, and operation logs from the MAE and analyzes the fault. After an analysis is complete, the fault analysis results are provided to help users identify potential causes and accelerate service recovery. LTE wireless fault management provides the following functions:

- **Fault overview**
Multisite faults are analyzed to help users locate the faulty base stations or cells. Users can continue to perform fast fault diagnosis or information collection.
- **Fast fault diagnosis**
After the faulty base stations or cells are determined using the fault overview function or manually selected, fault diagnosis can be performed on a single faulty base station or cell.
- **Information collection**
After the faulty base stations are determined using the fault overview function or manually selected, information about a single faulty base station can be collected.
- **All scenario KPI overview**
This function analyzes KPI trends of multiple sites in all scenarios and displays the results.
- **Fault recovery confirmation**
Visualized graphics are provided to display changes such as KPI changes and cell status changes of NEs after the recovery plan is executed.
- **Automatic fault detection**
It automatically detects cell faults and some KPI faults for fault diagnosis and information collection.

Figure 3-7 shows the relationships between the preceding functions.

Figure 3-7 Relationships between LTE fault management functions

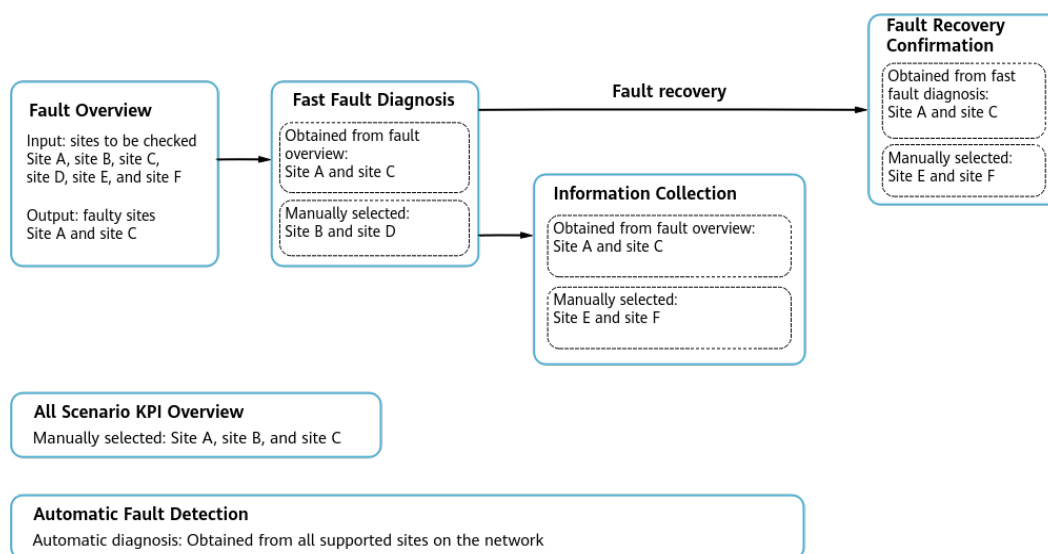


Table 3-17 lists the supported LTE wireless fault scenarios.

Table 3-17 LTE wireless fault scenarios

Scenario	Sub-Scenario	Check Item
Performance KPI (FDD/TDD)	Paging	Paging Messages Discarded due to PchCong
		Paging Messages Discarded from the MME
	Access	RRC Success Rate
		S1SIG Setup Success Rate
		E-RAB Setup Success Rate
		RACH Setup Success Rate
	Service Drop	Call Drop Rate
	Data Transmission	UL User Throughput(Kbit/s)
		UL Cell Throughput(Kbit/s)
		UL Cell Traffic Volume(bit)
		DL User Throughput(Kbit/s)
		DL Cell Throughput(Kbit/s)
		DL Cell Traffic Volume(bit)
	IRAT	CSFB Execution Success Rate
		CSFB Preparation Success Rate
		IRAT PSHO Execution Success Rate
		IRAT PSHO Preparation Success Rate
		IRAT SRVCC Execution Success Rate
		IRAT SRVCC Preparation Success Rate
		CSFB To 3G Ratio
	Handover	Inter-Frequency HO Success Rate
		Intra-Frequency HO Success Rate
		Handover In Success Rate
		Handover In Pre Success Rate
		X2 HO Out Success Rate
		S1 HO Out Success Rate
		FDD TDD HO In Success Rate

Scenario	Sub-Scenario	Check Item
		FDD TDD HO Out Success Rate
		FDD TDD HO In Pre Success Rate
		FDD TDD HO Out Pre Success Rate
	Voice	E-RAB Setup Success Rate QCI1
		E-RAB Setup Success Rate QCI5
		Call Drop Rate QCI1
		Call Drop Rate QCI5
		UL Packet Loss Rate QCI1
		DL Packet Loss Rate QCI1
		DL PDCP Packet Loss Rate QCI1
		Consecutively Lost Downlink Packet Statistics
		DL Low Quality Calls Rate
		UL Low Quality Calls Rate
		DL Silence Calls Rate
		UL Silence Calls Rate
		DL Poor and Bad VQI Rate
		UL Poor and Bad VQI Rate
		E2E Poor and Bad VQI Rate
Performance KPI (NB-IoT)	Access	NB-IoT RRC Success Rate
		NB-IoT User Number
		NB-IoT RACH Success Rate
	Service Drop	NB-IoT Call Drop Rate
	Data Transmission	NB-IoT DL Packet Loss Rate(SRB)
		NB-IoT UL RBLER
		NB-IoT DL RBLER
Performance KPI (eMTC)	Access	eMTC RACH Setup Success Rate
		eMTC RRC Setup Success Rate
	Service Drop	eMTC Call Drop Rate
	Data Transmission	eMTC DL IBLER

Scenario	Sub-Scenario	Check Item
		eMTC UL IBLER
		eMTC DL RBLER
		eMTC UL RBLER
	Handover	eMTC Inter-Frequency HO Success Rate
		eMTC Intra-Frequency HO Success Rate
		eMTC Handover In Success Rate
Cell Fault	Cell Unavailable	Cell Unavailable
	Cell Capability Degraded	Cell Capability Degraded
	RF Out of Service	RF Out of Service
Heavy Traffic	Heavy Traffic	Heavy Traffic
Interference	Interference	TDD UL Interference
	1588 V2 Clock Source Fault	1588 V2 Clock Source Fault
NSA Networking	Change in the Number of NSA UEs	Average Number of NSA UEs
		Success Rate of Outgoing Handovers for NSA PCC Anchoring
		Success Rate of Handover Executions for NSA PCC Anchoring
		Success Rate of Handover Preparations for NSA PCC Anchoring
	NR B1 Measurement	Number of NR B1 Measurements
		B1 Measurement Success Rate for NSA UEs
	SCG Addition	SgNB Addition Success Rate
	E-RAB Change	NSA User-Plane Bearer Change Success Rate
	Abnormal SCG Release	Abnormal SgNB Release Rate
		SgNB Failure Rate
	Service Drop	Service Drop Rate of NSA UEs
		Service Drop Rate of Non-NSA UEs

Scenario	Sub-Scenario	Check Item
	Handover	Success Rate of Outgoing Handovers for NSA UEs
		NSA DC Pcell Change Success Rate
	SCG Change	SgNB Change Success Rate
		LTE-triggered SCG Change Success Rate
		NR-triggered SCG Change Success Rate
	Abnormal Traffic Volume	Total PDCP-layer Traffic Volume of Downlink Distributed Data Transmitted for NSA UEs in a Cell(bit)
		Total PDCP-layer Traffic Volume of Uplink Distributed Data Received for NSA UEs in a Cell(bit)

Except heavy traffic, all other scenarios support fault overview and fast fault diagnosis. Heavy traffic generally occurs only in a small number of sites, so users are advised to select heavy-traffic sites for a fast fault diagnosis.

For details of how to use the FMA to handle LTE wireless faults, see **Fault Management** > **FMA** in *iMaster MAE Product Documentation*.

3.1.2.2.1 Fault Overview

Fault overview provides an overview of overall faults to help users learn the entire network's operating status.

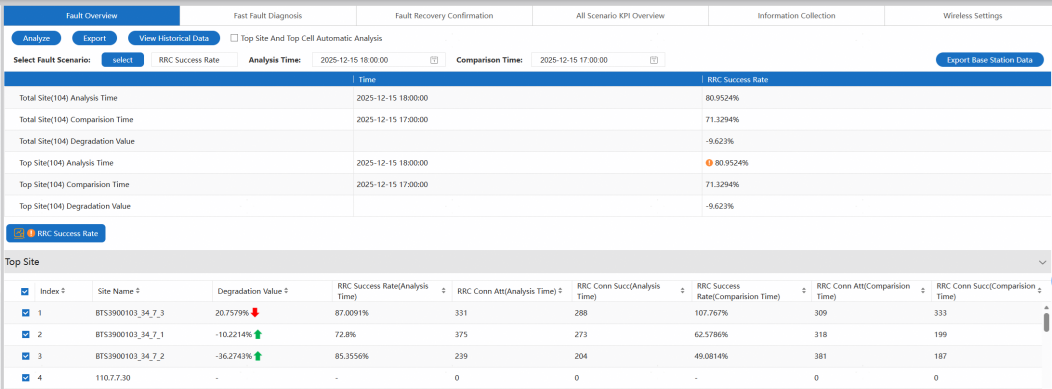
After users select fault scenarios (as listed in [Table 3-17](#)), LTE sites to be analyzed, the fault occurrence time, and the comparison time, the fault overview function performs fault analysis. Fault overview provides different information for performance KPI, cell, heavy-traffic, NSA networking, and interference faults of base stations.

Performance KPI Faults

For performance KPI faults of base stations in LTE-only or NSA networking, fault overview displays top sites, KPI trend chart, KPI deterioration time, distribution of KPI failure cause values and comparison with those in the same period of the last week, and the site list.

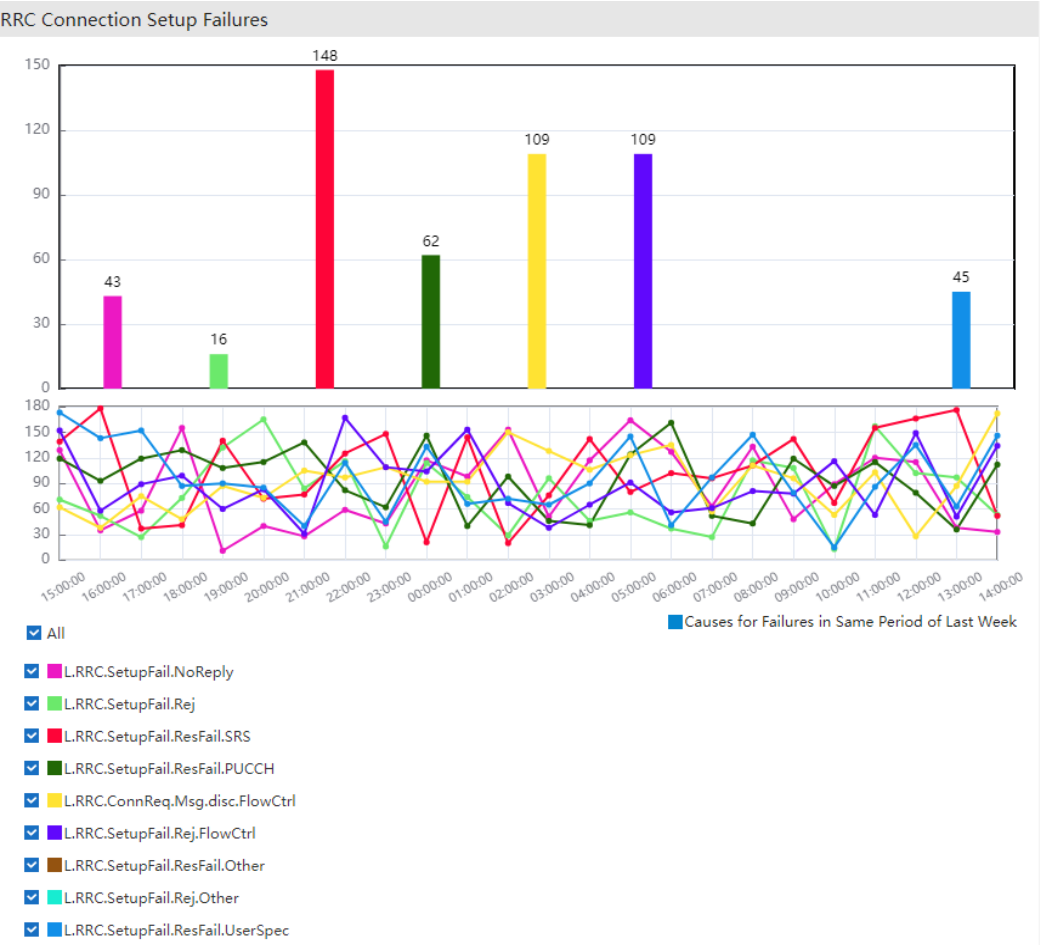
- Top site
Includes site name, KPIs obtained on the fault occurrence day, and KPIs obtained at the comparison time, as shown in [Figure 3-8](#).

Figure 3-8 Top site example



- Trend analysis
Displays the KPI trends during the 24 hours before the fault occurs, the past 48 hours to the past 24 hours, and the same period on the same day of the last week.
- KPI deterioration time
Displayed on a KPI trend chart.
- Distribution of KPI failure cause values and comparison with those at the same time of the last week
Displays the values of the counters used to calculate a KPI at the fault occurrence time and at the same time of the last week, as shown in [Figure 3-9](#).

Figure 3-9 Failure cause value distribution example



- Site list
 - For traditional base stations that require further analysis, fast fault diagnosis or information collection can be performed.

Figure 3-10 Site list example (for traditional base stations)

Site List		
NEs to Be Analyzed:3		
☒	Site Name	Contribution Rate
☒	LTE_100_034_089	0%
☒	LTE_100_034_077	0%
☒	LTE_100_034_078	0%
☐	LTE_100_034_079	0%
☐	LTE_100_034_080	0%
☐	LTE_100_034_081	0%
☐	LTE_100_034_082	0%
☐	LTE_100_034_083	0%
☐	LTE_100_034_084	0%
☐	LTE_100_034_085	0%
☐	LTE_100_034_086	0%

 NOTE

The FMA provides automatic analysis. The automatic analysis function applies to fault scenarios that are easily identified. If the KPIs affected by a fault are not easily identified or multiple KPIs need to be analyzed, automatic analysis is not recommended.

- If **Automatic Analysis** is selected, the FMA automatically performs fault overview and fast fault diagnosis after you click **Analyze**. No manual intervention is required.
- If **Automatic Analysis** is not selected, the FMA only performs fault overview after you click **Analyze**.

Cell Faults

For cell faults, fault overview displays the following information:

- According to the selected faulty scenario, the numbers of sites and cells where the fault occurs are displayed. The alarms are classified into active alarms and historical alarms. [Figure 3-11](#) shows an example.

Figure 3-11 Example of the number of sites and cells with faults

Items	Cell Unavailable		eNodeB Out of Service	
	Current Alarm	History Alarm	Current Alarm	History Alarm
Site Num	6	0	0	0
Cell Num	7	0	0	0

 NOTE

Only historical alarms generated within 24 hours before the analysis are recorded.

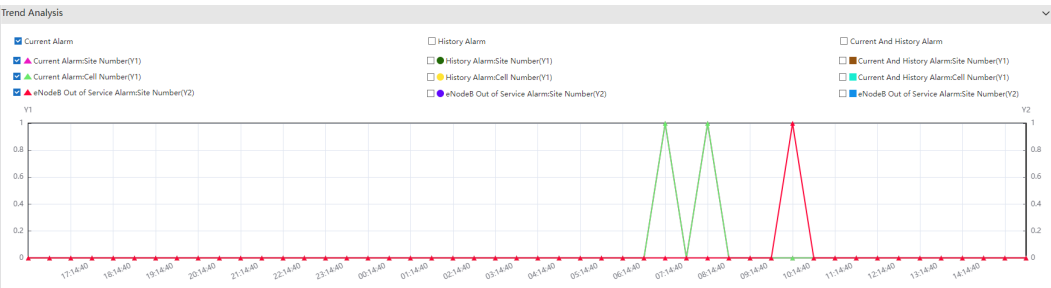
- Fault cell list: displays the number of sites and cells where an alarm related to the fault is reported. The alarms are classified into active alarms and historical alarms. [Figure 3-12](#) shows an example.

Figure 3-12 Example of the fault cell list

Fault Cell List				
Current Alarm				
History Alarm				
4/4				
☒	Index	Alarm Details	Site Num	Cell Num
☒	1	RF module abnormal	2	5
☒				
☒	Index	eNodeBID	Site Name	Cell Num
☒	1	22321	zqp_144	3
☒	2	173222	KZ_CoMPT_LNR_11.117	2
Total records: 2				
			5/page	
☒	2	Insufficient license	1	1
☒	3	The BBP capability is insufficient	1	1
Total records: 3				
			5/page	
Refresh Statistical Trend				

- Alarm quantity change trend. [Figure 3-13](#) shows an example.

Figure 3-13 Alarm quantity change trend example



Interference Faults

The **Fault Overview** window displays TDD uplink interference, if a user selects the interference fault scenario.

- Automatic analysis
Not supported
- Export base station data
Exports data, such as the NE name, local cell ID, degradation value, analysis time, and comparison time, about all base stations of the subnet that is being analyzed. [Figure 3-14](#) shows an example.

Figure 3-14 Base station data example

NE Name	TDD UL Interference(Degradation)	TDD UL Interference(2023-07-03 14:00:00)	TDD UL Interference(2023-07-02 14:00:00)
4GFMA:LocalCellId:2	-30	49	79
4GFMA:LocalCellId:3	-13	64	77

- Site list
Displays the contribution rate of TDD cells served by each eNodeB in the current eNodeB group to KPI failures or deterioration within the performance measurement period starting from *T0*. Sorts the eNodeBs in descending order of the contribution rate. [Figure 3-15](#) shows an example.

NOTE

Contribution rate to TDD uplink interference KPI faults = (Absolute value of the difference in a KPI value at the same time of the last week and that at the current time in a cell/Sum of the absolute values of the differences in the KPI value at the same time of the last week and that at the current time in all cells) x 100%

Figure 3-15 Site list example

Site List	
Site Name	Contribution Rate
4G_Interference_2_LocalCellId:67	100%

- Top site
Indicates KPIs of top *N* sites. The number of top *N* sites is set to 100 by default. All TDD cells of top 100 sites are displayed. If the total number of sites on the subnet or in the subzone does not exceed 100, KPI values of the subnet are the same as those of top *N* sites. [Figure 3-16](#) shows an example.

 NOTE

Top *N* sites are filtered by the following principle: compare KPI values at the KPI deterioration time and the specified time for all sites on the subnet or in the subzone, and select top 100 sites with the maximum KPI deterioration.

Figure 3-16 Top site list

Top Site				
☒ Index	Site Name	Degradation Value(dB)	TDD UL Interference(dBm)(Analysis Time)	TDD UL Interference(dBm)(Comparison Time)
☒ 1	4G_Interference_2_LocalCellId67	-4	-114	-110
Refresh Statistical Trend Fault Reanalysis				

- High-risk interference source site
Summarizes suspicious sites identified based on analysis results of top 3 sites, clock alarms, forcible phase synchronization events, and DA traffic statistics. [Figure 3-17](#) shows an example.

 NOTE

Sites where events have occurred are listed first, followed by those with alarms and other sites.

Figure 3-17 High-risk interference source site example

High-risk Interference Source Site					
TOPN	Site	Alarm	Event	DA Value	TopSite
1	S101--00617130	FALSE	FALSE	FALSE	TRUE
2	BTS3900101_10_43_1	FALSE	FALSE	FALSE	TRUE
3	BTS3900101_10_43_11	FALSE	FALSE	FALSE	TRUE
Total records: 3					

- Symbol-level interference level
Displays the values of all symbol-level interference KPIs in the current eNodeB group during the performance measurement period starting from *T0*. [Figure 3-18](#) shows an example.

 NOTE

T0 is the start time of the first performance measurement period with KPI values changing from stable to dramatic.

Figure 3-18 Symbol-level interference level example



3.1.2.2.2 Fast Fault Diagnosis

Fast fault diagnosis analyzes the performance of top sites identified using the fault overview function. The analysis result provides associated information about KPIs, active alarms, and operation logs for each of the top sites and each cell served by the top sites at the time when KPIs deteriorate. By analyzing KPIs, active alarms, and operation logs, this function helps users identify the minimum faulty units.

For traditional base stations, fast fault diagnosis displays different diagnosis information for performance KPI, cell, heavy-traffic, interference, and NSA networking faults.

For the detailed procedure of fast fault diagnosis, see "Starting Fast Fault Diagnosis" in *iMaster MAE Product Documentation*.

Performance KPI Faults

For performance KPI and NSA networking faults, the fast fault diagnosis function provides site- and cell-level diagnosis information, as described in [Table 3-18](#) and [Table 3-19](#).

Table 3-18 Site-level diagnosis information for performance KPI faults

Function	Description
List of Common Issues	Collects information about sites with updated version, license-related alarm, or high-risk operations, and displays result in the form of "exist or not (existing quantity)". "Exist or not" represents YES or NO. "Existing quantity" represents the number of sites with updated version, license-related alarm, or high-risk operations. If the information of "exist or not" is YES, you can click to view the detailed information, and the sites with a YES answer are listed before those with a NO answer in the basic site information table.
Basic Site Information Table	• Site Name indicates the name of a site. • Contribution Rate indicates the ratio of failures in a site to the total failures, which is analyzed by the fast fault diagnosis function at a specified time. • Actual value of a KPI indicates a site-level KPI value generated when the KPI is analyzed. • Denominator (number of attempts) of a KPI indicates the number of setup attempts related to this KPI, if a non-traffic-volume-related KPI (such as success rate or call drop rate) is analyzed. • Version Change Info lists the latest software version updates of a site. • License Analysis displays active license-related alarms at the fault occurrence time. • Risky Operation displays risky operation records in a specified period.

Function	Description
Transmission Diagnosis Info	The following KPIs involve transmission diagnosis and analysis: • RRC Success Rate • eMTC RRC Setup Success Rate • S1SIG Setup Success Rate • E-RAB Setup Success Rate • E-RAB Setup Success Rate QCI1 • E-RAB Setup Success Rate QCI5 • CSFB Execution Success Rate • X2 HO Out Success Rate • FDD TDD HO Out Pre Success Rate Transmission diagnosis information is provided on the Original Messages and Diagnosis Results tab pages. The following eNodeB transmission diagnosis information is displayed: • Transmission port information (the DSP ETHPORT command output) • Address Resolution Protocol (ARP) information learned by the transmission ports (the DSP ARP command output) • Configured route information (the output of the following commands) – DSP IPRT when the old transmission configuration model is used – DSP IPROUTE4 when the new transmission configuration model is used • S1 interface transmission quality information (the output of the PING command over the S1 interface)
Active Alarms	Displays the active alarms of the to-be-analyzed site. Known alarms on specific sites can be used to identify other sites with the same alarms. Statistics on active alarms in a time segment are collected, and the alarm distribution is displayed.
Abnormal Voice CHR Event Analysis	Collects statistics about abnormal voice events of all cells of the current site including DL Lost Words, DL VOLTE PICP Abnormal, DL VQI Deterioration, DL VQI VOICE UU BURST PKTLOSS, UL Lost Words, UL VOLTE PICP Abnormal and UL VQI Deterioration, and displays the statistics in a summary table. You can click a value in the CauseTimes column to view the details about the corresponding exception name and cause. You can filter cell IDs in the details table to view detailed information about exceptions of a single cell. This function takes effect only in the voice sub-scenario.
Fault Tree	Displays initial analysis results of the site. eMTC KPIs are not included.

Function	Description
Voice Parameters	This function takes effect only for voice services. - Identifies voice-related core parameters of a specified site, compares the parameter settings of the current eNodeB and baseline parameter settings, and displays comparison results. - Allows users to import voice baseline parameter files and perform simple checks.
Access Parameters	This function takes effect only for access services. (eMTC counters are not included.) Identifies access-related core parameters of a specified site, compares the parameter settings of the current eNodeB and baseline parameter settings, and displays comparison results.
Call drop parameters	This function applies only to the call drop sub-scenario. (eMTC counters are not included.) Identifies call drop-related core parameters of a specified site, compares the parameter settings of the current eNodeB and baseline parameter settings, and displays comparison results.
Data Transmission Parameters	Identifies data-transmission-related core parameters of a specified site, and compares the parameter settings of the current eNodeB with baseline parameter settings. This function takes effect only in the NB-IoT data transmission sub-scenario.

 NOTE

On the **Site** tab page for **KPI Fault Scenario**, only the basic site information table, active alarms, and parameter check will be displayed for NB-IoT counters (with NB-IoT RRC Success Rate, NB-IoT RACH Success Rate, and NB-IoT Call Drop Rate being exceptions). The KPI fault tree is not provided on the **Cell** tab page.

Table 3-19 Cell-level diagnosis information for performance KPI faults

Function	Description
List of Common Issues	Collects statistics on whether the cells in the basic cell information table are heavily loaded or affected by TDD interference, and provides a conclusion in the form of "exist or not (existing quantity)". "Exist or not" represents YES or NO. "Existing quantity" represents the number of cells that are heavily loaded or affected by TDD interference. If the information of "exist or not" is YES, you may click to view the detailed information, and the cells with a YES answer are listed before those with a NO answer in the basic cell information table.

Function	Description
Basic Cell Information Table	• LocalCellId indicates the cell ID. • Contribution Rate indicates the ratio of failures in a cell to the total failures, which is analyzed by the fast fault diagnosis function at a specified time. • Actual value of a KPI indicates the cell KPI value at the time when the KPI is analyzed. • Denominator (number of attempts) of a KPI indicates the number of setup attempts related to this KPI, if a non-traffic-volume-related KPI (such as success rate or call drop rate) is analyzed. • Heavily Loaded determines whether a cell is highly loaded based on the CPU usage of the board serving the cell and the number of online users in the cell. • TDD Interference: For TDD cells, the FMA determines whether the following interference types exist based on performance counters related to uplink interference: – Clock out-of-synchronization interference or long-distance interference – Clock out-of-synchronization interference – Other interference If the preceding interference exists, a specific interference type is displayed. If no interference-related performance counters are obtained, "-" is displayed. If no interference exists, "No interference" is displayed. For FDD cells, the FMA does not analyze TDD interference and "Not TDD cell" is displayed.
Coverage Level NOTE The Cell tab page shows the coverage level only when it is used for analyzing NB-IoT KPIs (except NB-IoT DL Packet Loss Rate(SRB)) in fault scenarios.	• Coverage level list – Coverage Level ID indicates the coverage level ID. – Contribution Rate indicates the ratio of failures of each coverage level in a cell to the total failures, which is analyzed by the fast fault diagnosis function at a specified time. – Actual value of a KPI indicates the cell coverage level at the time when the KPI is analyzed. – Denominator (number of attempts) of a KPI indicates the number of setup attempts related to the coverage level, if a non-traffic-volume-related KPI (success rate or call drop rate) is analyzed. • Coverage level trend Provides coverage level trend during an analysis period.

Function	Description
Cell Dashboard	Provides correlation analysis on the cell KPIs, alarms, and operation logs. After users click a time point when a KPI changes, the alarms and operation logs generated at the time will be displayed to help users quickly find the potential fault cause. Users can select records generated in the latest two hours before the fault occurs from operation logs for MAE-Deployment analysis. For cell faults, select records generated at the current time. The MAE-Deployment can only analyze records generated within two hours before the fault occurrence time, and can only analyze MOD, RMV, and ADD commands.
Board Diagnosis Info	Displays the CPU usage trends of the main control board and baseband processing unit (BBP) serving a cell and the trend of the number of connected cell users. This helps users observe the relationship between the cell KPI changes and the board CPU usage.
Abnormal Voice CHR Event Analysis	Collects statistics about abnormal voice events of the current cell including DL Lost Words, DL VOLTE PICP Abnormal, DL VQI Deterioration, DL VQI VOICE UU BURST PKTLOSS, UL Lost Words, UL VOLTE PICP Abnormal and UL VQI Deterioration, and displays the statistics in a summary table. You can click a value in the CauseTimes column to view the details about the corresponding exception name and cause. This function takes effect only in the voice sub-scenario.
Cell KPI Analysis	Displays the trends of multiple cell-level counters. eMTC-related counters are not involved.
Fault Tree	Displays the preliminary fault analysis result on the current cell. The result includes the diagnosis result summary and detailed diagnosis data of each sub-scenario in the fault tree. eMTC counters are not included.

Cell Faults

For cell faults, the fast fault diagnosis function provides only site-level diagnosis information, as described in [Table 3-20](#).

Table 3-20 Diagnosis information for cell faults

Function	Description
Site Key Configuration Table	Lists information about each faulty cell served by the site, including: • Basic configuration: Cell FDD TDD indication, NB-IoT Cell Flag, Downlink EARFCN, Downlink bandwidth, Cell transmission and reception mode, High speed flag, Subframe assignment, Special subframe patterns, and others • Operator information: Local tracking area ID, CN Operator ID, Tracking area code, CN Operator name, CN Operator type, Mobile country code, Mobile network code, and others • S1 interface information: S1 Interface ID, S1 Interface State, S1 Interface Is Blocked, S1 Interface Fault Reason Provides the functions of analyzing and exporting power information. • Analyze Power Information: Analyzes power information for the cell unavailability indicator. • Export Power Information: Exports the analysis results of power information, including basic cell configuration information and power information of each channel of the faulty cell.
Cell Device Info	Includes the numbers of sector devices, channels, baseband processing boards, and carriers on each TX channel.
Cell Dashboard	Lists the following information about a faulty cell: • Counters related to cell unavailability, including L.Cell.Unavail.Dur.Sys, L.Cell.Unavail.Dur.Manual, L.Cell.Unavail.Dur.EnergySaving, and L.Cell.Unavail.Dur.Sys.S1Fail • Cell alarms • Cell operation logs (excluding those related to DSP and LST MML commands)
Fault Diagnosis Result of Cells	Displays the DSP CELL command output for cell status analysis, and provides cell setup failure causes based on cell unavailability error information. There are no diagnosis results indicating the degrading of cell serving capability and no diagnosis result indicating RF unit service unavailability.

Heavy-Traffic Faults

For heavy-traffic faults, the fast fault diagnosis function provides site- and cell-level diagnosis information, as described in [Table 3-21](#) and [Table 3-22](#).

Table 3-21 Site-level diagnosis information for heavy-traffic faults

Function	Description
Basic Site Information Table	• Site Name indicates the name of a site. • RRC Success Rate indicates the site-level RRC setup success rate. • E-RAB Setup Success Rate indicates the site-level E-RAB setup success rate. • Call Drop Rate indicates the site-level call drop rate. • DL Cell Throughput (Kbit/s) indicates the site-level downlink cell throughput. • UL Cell Throughput (Kbit/s) indicates the site-level uplink cell throughput.
Active Alarms	Displays the current active alarms of the to-be-analyzed site. Known alarms on specific sites can be used to identify other sites with the same alarms.
Fault Diagnosis Result of Sites	Displays the preliminary fault analysis result on the current site and provides recovery suggestions.

Table 3-22 Cell-level diagnosis information for heavy-traffic faults

Function	Description
Basic Cell Information Table	• LocalCellId indicates the cell ID. • RRC Success Rate indicates the cell-level RRC setup success rate. • E-RAB Setup Success Rate indicates the cell-level E-RAB setup success rate. • Call Drop Rate indicates the cell-level call drop rate. • DL Cell Throughput (Kbit/s) indicates the cell-level downlink cell throughput. • UL Cell Throughput (Kbit/s) indicates the cell-level uplink cell throughput.
Cell Dashboard	Provides correlation analysis on the cell KPIs, alarms, and operation logs. After users click a time point when a KPI changes, the alarms and operation logs generated at the time will be displayed to help users quickly find the potential fault cause.
Fault Diagnosis Result of Cells	Displays the preliminary fault analysis result on the current cell and provides recovery suggestions.

Interference Faults

If TDD interference fault scenarios are selected in fault overview, and **Next** is clicked to start fast fault diagnosis, only site-level diagnosis information is displayed, as described in [Table 3-23](#).

Table 3-23 Site-level diagnosis information for interference faults

Function	Description
Clock Alarm Analysis	Displays analysis results of clock alarms for TDD sites on the entire network.
Forcible Phase Synchronization Event Analysis	Displays analysis results of forcible phase synchronization events for TDD sites on the entire network.
Risky Operation	Provides the following two types of information: - Information of the TDD sites with the following MML command operation logs on the entire network: ACT CELL MOD CELLFRAMEOFFSET ADD CELLFRAMEOFFSET RMV CELLFRAMEOFFSET MOD RRU MOD CELL - Information of TDD sites with EVT-26213 NE Startup reported.
DA Traffic Statistics	Displays analysis results of DA traffic statistics that meet the conditions of all TDD sites on the network.
Distribution of Symbol-Level Interference Characteristics	Provides a distribution of interference characteristics according to the diagnosis logic, using all TDD cells at all sites based on the analysis results of clock alarms, forcible phase synchronization events, risky operations, and DA traffic statistics.

Descriptions of Fault Tree and Diagnosis Report

The fault tree displays the complete diagnosis results, as shown in [Figure 3-19](#). Diagnosis items highlighted in red indicate faults which can be linked to detailed diagnosis results in the sub-scenario. Diagnosis items highlighted in green indicate that no faults occur and no detailed results are provided.

Figure 3-19 Diagnosis result example

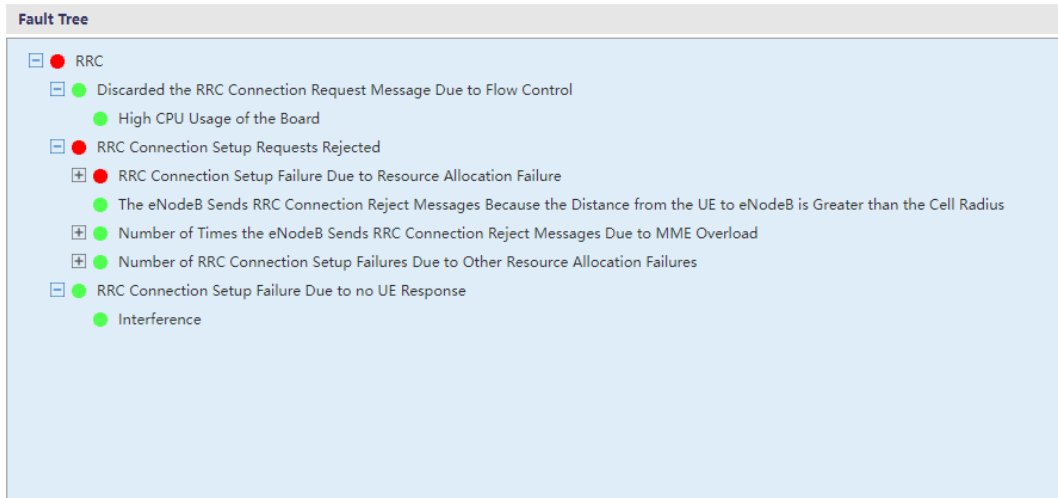
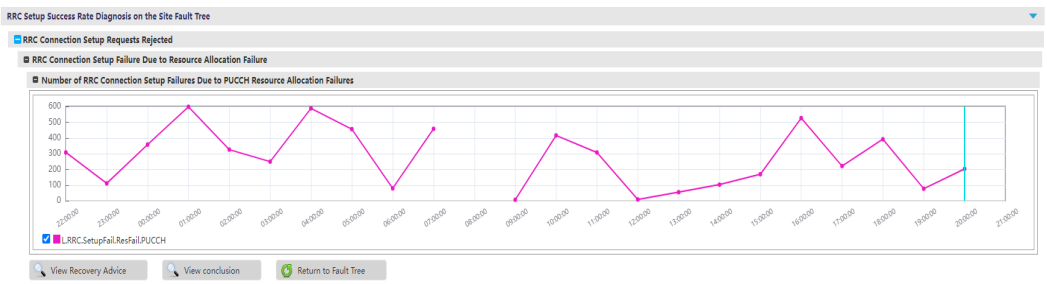


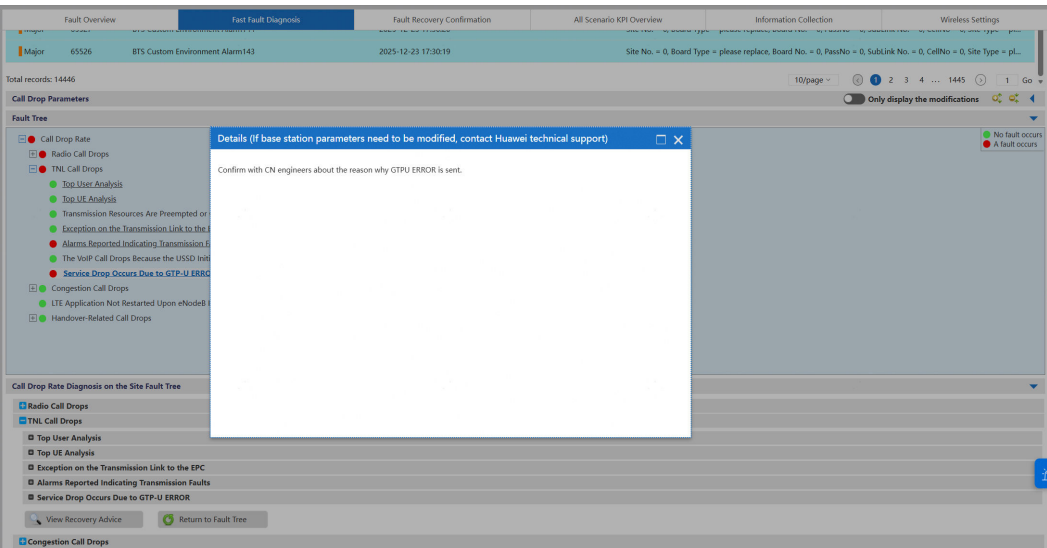
Figure 3-20 shows detailed diagnosis results in a sub-scenario.

Figure 3-20 Detailed diagnosis result example in a sub-scenario



Corresponding recovery advice is provided as a reference after fault diagnosis in a sub-scenario. For details, see [Figure 3-21](#).

Figure 3-21 Recovery advice example in a sub-scenario



Imported Voice Parameter Check

The FMA allows users to import basic voice parameters on the GUI and perform simple parameter check on imported parameters without conditions. This function is controlled by a switch, and the voice parameter check can be performed only when the switch is turned on. The voice parameter check tab page and the related descriptions are shown in [Figure 3-22](#) and [Table 3-24](#).

Figure 3-22 Voice parameter check tab page

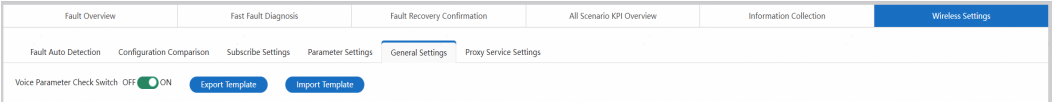


Table 3-24 Description of the voice parameter check tab page

Function	Description
Voice Parameter Check Switch	• If the switch is turned off, voice baseline parameter files cannot be imported, and only traditional voice parameter check is supported. • If this switch is turned on, voice baseline is imported for parameter check. By default, this switch is turned on.
Export Template	Export the baseline template for voice parameter check.
Import Template	Import the baseline template for voice parameter check. Only file types of .xlsx and .xls are supported.

Users need to turn on the voice parameter check switch, export a copy of basic voice parameters via exporting the voice parameter template, and then modify the exported voice parameters. Click **Import Template** to import the voice parameter template. The FMA automatically checks the voice parameters in the imported template. Start fast fault diagnosis and view the check result in the **Voice Parameters** area. For details, see [Performance KPI Faults](#).

3.1.2.2.3 Information Collection

This function can quickly and accurately collect onsite information and improve fault recovery efficiency.

- An information collection task involves the LTE, NB-IoT, eMTC, RanSharing, 5G-relation, and AutoDetect scenarios. By default, fault information in LTE scenarios is preferentially collected and that in other scenarios is collected in sequence. To preferentially collect information in a specific scenario, select **LTE first**, **NB-IoT first**, **eMTC First**, or **RanSharing first** when creating a task.
- If the site is an NSA site, the network information collection (NIC) tool collects the first-batch data on the NR side by default.
- Related information and operations are provided in the following columns for a created fault information collection task: **Task Name**, **Progress**, **Status**,

Sources, Execution Time, Run Time, Created On, Created By, and Operation. The value of **Sources** can be **Manual** or **Auto Detect**. For manually created tasks, the **Details**, **Stop**, and **Delete** buttons in the **Operation** column can be used to query task details, stop the task, and delete the task, respectively. A stopped task can be deleted but cannot be resumed. For automatic detection tasks, only the **Download** button is available in the **Operation** column.

- The automatic detection results collected by the NIC can be displayed.

Table 3-25 lists the collected fault information.

Table 3-25 Collected fault information

Collection Item	Scope
NE configuration file	One file per NE
MML query output	One file per NE
Results of pinging a base station from the MAE	One file per NE
Results of tracing routes from the MAE to a base station	One file per NE
Performance indicators	During a specified collection period
Alarm data	During a specified collection period
Event data	During a specified collection period
Operation logs saved on the MAE	During a specified collection period
Original operation logs of top NEs	One file per NE
Automatic detection report (only involved in manually created information collection tasks)	Consisting of reports of Network-wide Automatic Detection in All Scenarios and Cell Unavailable Automatic Detection within seven days (A maximum of 100 MB reports can be collected.)

For details, see "Network Information Collection" in *iMaster MAE Product Documentation*.

3.1.2.2.4 All Scenario KPI Overview

This function supports site-level KPI and performance counter analysis for NEs managed by the MAE. The supported scenarios are **KPI** and **MOCN**. Specifically,

- Users can select NEs from the network topology or select a version group on the **All Scenario KPI Overview** tab page for analysis. When the KPI period is set to **P60**, a maximum of 2000 NEs can be analyzed at a time. For other KPI periods, the maximum number of NEs that can be analyzed at a time is

displayed on the GUI. It is recommended that no more than one user use the all scenario KPI overview function at the same time. In addition, when using this function, you are not advised to use other fault analysis functions.

- Users can select the analysis date (the current day is selected by default). The analysis period is [00:00, 24:00), which is fixed to 24 hours. After the analysis is complete, the KPI statistics of the current day (corresponding to **today**), the previous day (corresponding to **yesterday**), and the same day of the previous week (corresponding to **last week**) are displayed by default for comparison.
- Users can specify a KPI period to view the statistical data in different KPI periods. The supported values on the GUI are **P5**, **P15**, **P30**, and **P60**, which correspond to 5 minutes, 15 minutes, 30 minutes, and 60 minutes, respectively. The default value is **P60**. Users need to ensure that the selected KPI period has been subscribed to.
- Users can export the analysis results. If the selected NEs are changed, the data analyzed in the current session of all scenario KPI overview can be exported using fault overview, fast fault diagnosis, and fault recovery confirmation.

Table 3-26 KPIs or performance counters applicable to LTE

Scenario	KPI/Performance Counter
KPI	RRC Success Rate
	E-RAB Setup Success Rate
	S1SIG Setup Success Rate
	Inter-Frequency HO Success Rate
	Intra-Frequency HO Success Rate
	E2E Handover In Success Rate
	Call Drop Rate
	DL Cell Traffic Volume(bit)
	Average User Number
	CSFB Success Rate
	SgNB Addition Success Rate
	SgNB Change Success Rate
	LTE-triggered SCG Change Success Rate
	Service Drop Rate of NSA UEs
	Average Uplink Interference
MOCN	E-RAB Setup Success Rate PLMN
	VoLTE E-RAB Setup Success Rate PLMN
	Call Drop Rate PLMN

Scenario	KPI/Performance Counter
	Call Drop Rate PLMN QCI1
	DL Traffic PLMN(MB)
	UL Traffic PLMN(MB)

3.1.2.2.5 Fault Recovery Confirmation

The FMA supports fault recovery confirmation in fast fault recovery scenarios. It showcases the changes in KPIs or cell status after the recovery plan is executed. The changes of NE KPIs, cell status, and alarm status are displayed in charts. Fault handling personnel can use such data to determine whether a fault is rectified. [Table 3-27](#) lists the detailed functions of fault recovery confirmation.

Table 3-27 Functions of fault recovery confirmation

Function	Sub-Function	Description
Fault recovery confirmation	1-minute KPI trend	Displays the 1-minute KPI trend of the faulty site after the recovery plan is executed, helping users to determine whether the fault is resolved.
	List of alarm clearance status	Displays the alarm clearance status after the recovery plan is executed, helping users to determine whether the fault is resolved.
	List of cell status	Displays the status of the cell in the fault scenario after the recovery plan is executed, helping users to determine whether the fault is resolved.

3.1.2.2.6 Automatic Fault Detection

Network-wide Automatic Detection in All Scenarios

The FMA supports periodic subscription to KPIs and automatic KPI analysis through the MAE-Access. When one or multiple associated KPIs on the network are abnormal and the number of abnormal sites reaches a certain value, the FMA can automatically identify and diagnose the exceptions, generating an analysis report for quick review. The report provides KPI analysis, alarm-related fault analysis, analysis of impact scope, and analysis related to operation and configuration, helping users to quickly rectify faults. The network-wide automatic detection in all scenarios function is enabled by default.

The information collection association switch is turned on by default. After the switch is turned on, reports of automatic fault detection are automatically packaged for the information collection function.

The automatic detection information collection switch is turned on by default. After the switch is turned on, automatic fault detection tasks collect fault information about faulty NEs.

You can set the thresholds for different scenarios on the **Scenario threshold** tab page. [Table 3-28](#) lists the details.

Table 3-28 Parameters related to automatic fault detection

Parameter	Description	Setting Notes
FMA fault analysis interval	Minimum interval for network-level fault analysis. The default value is 2 hours.	The smaller the value of this parameter, the more the network faults that can be determined within 24 hours. The larger the value of this parameter, the fewer the network faults that can be determined within 24 hours. The default value is recommended.
Level-1 entry threshold	Thresholds used to determine whether level-2 thresholds are required for further determination. If the KPIs of the current cell reach the level-1 entry thresholds, further analysis against level-2 thresholds is not required and analysis can be performed subsequently.	- For success rate-related KPIs, the larger the threshold values, the higher the probability of determining network faults. The smaller the threshold values, the lower the probability of determining network faults. - For deterioration rate-related KPIs, the larger the threshold values, the lower the probability of determining network faults. The smaller the threshold values, the higher the probability of determining network faults.

Parameter	Description	Setting Notes
Level-1 exit threshold	Thresholds used to determine whether the KPIs of the current cell are normal. If the exit thresholds are reached, the KPIs of the current cell are normal and no further analysis is required.	- For success rate-related KPIs, the larger the threshold values, the more the KPIs being judged against the level-2 thresholds. The smaller the threshold values, the fewer the KPIs being judged against the level-2 thresholds. - For deterioration rate-related KPIs, the larger the threshold values, the fewer the KPIs being judged against the level-2 thresholds. The smaller the threshold values, the more the KPIs being judged against the level-2 thresholds. - The more the KPIs being judged against the level-2 thresholds, the lower the analysis efficiency, and the higher the analysis accuracy.
Minimum sample size for level-1 entry threshold	Minimum numbers of samples used for determination against the level-1 entry thresholds, which prevents misjudgment caused by KPI value fluctuations when the number of UEs is small.	The larger the numbers, the stronger the anti-fluctuation capability. The smaller the numbers, the weaker the anti-fluctuation capability. The default values are recommended.

Parameter	Description	Setting Notes
Level-2 z-score entry threshold	Determines whether a KPI value fluctuation of the cell in the current period exceeds the level-2 entry threshold based on the KPI values in historical period. The level-2 threshold uses z-scores. If it exceeds the threshold, a fault is determined, and no subsequent analysis is required.	The larger the absolute values of the thresholds, the higher the tolerance to KPI value fluctuation. The smaller the absolute values of the thresholds, the lower the tolerance to KPI value fluctuation. The default values are recommended.
Minimum number of cells to determine a network fault	Minimum numbers of cells for determining a network fault.	The larger the numbers, the lower the probability of determining a network fault. The smaller the numbers, the higher the probability of determining a network fault. The default values are recommended.
Z-score entry threshold for a network fault	z-score thresholds for the numbers of cells. If the number of cells with a specific KPI fault reaches the corresponding threshold, the fault is considered as a network-level fault.	The larger the absolute values of the thresholds, the higher the tolerance to KPI value fluctuation. The smaller the absolute values of the thresholds, the lower the tolerance to KPI value fluctuation. The default values are recommended.

 NOTE

As adjusting the preceding thresholds may cause missing or excessive detection of network faults, adjust the thresholds under the guidance of Huawei engineers.

The FMA analyzes basic network KPIs every 60 minutes. If KPIs are abnormal for a large number of base stations, the FMA generates an auto-detection analysis report. [Table 3-29](#) lists the KPI scenarios defined by the MAE-Access.

Table 3-29 KPI scenarios defined for network-wide automatic detection in all scenarios

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
Common FDD/TDD Services	Paging	Base Station Paging Discard Rate	$(L.Paging.Dis.PchCong + L.Paging.Dis.PagingTimeout + L.Paging.Dis.Num)/L.Paging.S1.Rx \times 100\%$
	Access	RRC Success Rate	$L.RRC.ConnReq.Succ/L.RRC.ConnReq.Att \times 100\%$
		S1SIG Setup Success Rate	$L.S1Sig.ConnEst.Succ/L.S1Sig.ConnEst.Att \times 100\%$
		E-RAB Setup Success Rate	$L.E-RAB.SuccEst/L.E-RAB.AttEst \times 100\%$
		Number of Users	$L.Traffic.User.Avg$
	Service Drop	Call Drop Rate	$(L.E-RAB.AbnormRel.MME + L.E-RAB.AbnormRel)/(L.E-RAB.NormRel + L.E-RAB.AbnormRel) \times 100\%$
		Abnormal Service Drop Rate of a Base Station	$L.E-RAB.AbnormRel/(L.E-RAB.NormRel + L.E-RAB.AbnormRel) \times 100\%$
		Rate of Handover-caused Service Drops	$L.E-RAB.AbnormRel.HOFailure/(L.E-RAB.NormRel + L.E-RAB.AbnormRel) \times 100\%$

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
	Handover	E2E Outgoing Handover Success Rate of a Cell	$\frac{(L.HHO.IntraeNB.IntraFreq.ExecSuccOut + L.HHO.IntereNB.IntraFreq.ExecSuccOut + L.HHO.IntraeNB.InterFreq.ExecSuccOut + L.HHO.IntereNB.InterFreq.ExecSuccOut + L.HHO.IntraeNB.InterFddTdd.ExecSuccOut + L.HHO.IntereNB.InterFddTdd.ExecSuccOut) / (L.HHO.IntraeNB.IntraFreq.PrepAttOut + L.HHO.IntereNB.IntraFreq.PrepAttOut + L.HHO.IntraeNB.InterFreq.PrepAttOut + L.HHO.IntereNB.InterFreq.PrepAttOut + L.HHO.IntraeNB.InterFddTdd.PrepAttOut + L.HHO.IntereNB.InterFddTdd.PrepAttOut) \times 100\%}{}$
		E2E Incoming Handover Success Rate of a Cell	$\frac{(L.HHO.IntraeNB.ExecSuccIn + L.HHO.IntereNB.ExecSuccIn + L.HHO.InterFddTdd.IntraeNB.ExecSuccIn + L.HHO.InterFddTdd.IntereNB.ExecSuccIn) / (L.HHO.IntraeNB.PrepAttIn + L.HHO.IntereNB.PrepAttIn + L.HHO.InterFddTdd.IntraeNB.PrepAttIn + L.HHO.InterFddTdd.IntereNB.PrepAttIn) \times 100\%}{}$
		X2 HO Out Success Rate	$\frac{(L.HHO.X2.IntraFreq.ExecSuccOut + L.HHO.X2.InterFreq.ExecSuccOut + L.HHO.X2.InterFddTdd.ExecSuccOut) / (L.HHO.X2.IntraFreq.PrepAttOut + L.HHO.X2.InterFreq.PrepAttOut + L.HHO.X2.InterFddTdd.PrepAttOut) \times 100\%}{}$

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
		S1 HO Out Success Rate	$\frac{ \begin{aligned} & (L.HHO.InterNB.IntraFreq.ExecSuccOut + \\ & L.HHO.InterNB.InterFreq.ExecSuccOut - \\ & L.HHO.X2.IntraFreq.ExecSuccOut - \\ & L.HHO.X2.InterFreq.ExecSuccOut + \\ & L.HHO.InterNB.InterFddTdd.ExecSuccOut - \\ & L.HHO.X2.InterFddTdd.ExecSuccOut) / \\ & (L.HHO.InterNB.IntraFreq.PrepAttOut + \\ & L.HHO.InterNB.InterFreq.PrepAttOut - \\ & L.HHO.X2.InterFreq.PrepAttOut - \\ & L.HHO.X2.IntraFreq.PrepAttOut + \\ & L.HHO.InterNB.InterFddTdd.PrepAttOut - \\ & L.HHO.X2.InterFddTdd.PrepAttOut) \end{aligned} }{ } \times 100\%$
	IRAT	CSFB Execution Success Rate	$\frac{(L.CSFB.E2W + L.CSFB.E2G)}{(L.CSFB.PrepSucc + L.CSFB.LCS.PrepSucc)} \times 100\%$
		CSFB Preparation Success Rate	$\frac{(L.CSFB.PrepSucc + L.CSFB.LCS.PrepSucc)}{(L.CSFB.PrepAtt + L.CSFB.LCS.PrepAtt)} \times 100\%$
		IRAT PSHO HO Success Rate	$\frac{(L.IRATHO.E2W.ExecSuccOut + L.IRATHO.E2G.ExecSuccOut)}{(L.IRATHO.E2W.PrepAttOut + L.IRATHO.E2G.PrepAttOut)} \times 100\%$
		IRAT SRVCC Success Rate	$\frac{(L.IRATHO.SRVCC.E2W.ExecSuccOut + L.IRATHO.SRVCC.E2G.ExecSuccOut)}{(L.IRATHO.SRVCC.E2W.PrepAttOut + L.IRATHO.SRVCC.E2G.PrepAttOut)} \times 100\%$
	VoLTE	E-RAB Setup Success Rate QCI1	$L.E-RAB.SuccEst.QCI.1 / L.E-RAB.AttEst.QCI.1 \times 100\%$
		E-RAB Setup Success Rate QCI5	$L.E-RAB.SuccEst.QCI.5 / L.E-RAB.AttEst.QCI.5 \times 100\%$

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
		Call Drop Rate QCI1	$\frac{L.E-RAB.AbnormRel.QCI.1}{L.E-RAB.AbnormRel.QCI.1 + L.E-RAB.NormRel.QCI.1} \times 100\%$
		Call Drop Rate QCI5	$\frac{L.E-RAB.AbnormRel.QCI.5}{L.E-RAB.AbnormRel.QCI.5 + L.E-RAB.NormRel.QCI.5} \times 100\%$
		Uplink Air Interface Packet Loss Rate QCI1	$\frac{L.Traffic.UL.PktLoss.Loss.QCI.1}{L.Traffic.UL.PktLoss.Tot.QCI.1} \times 100\%$
		Downlink Air Interface Packet Loss Rate QCI1	$\frac{L.Traffic.DL.PktUuLoss.Loss.QCI.1}{L.Traffic.DL.PktUuLoss.Tot.QCI.1} \times 100\%$
		Downlink PDCP Packet Discard Rate	$\frac{L.PDCP.Tx.Disc.Trf.SDU.QCI.1}{(L.PDCP.Tx.Disc.Trf.SDU.QCI.1 + L.PDCP.Tx.TotRev.Trf.SDU.QCI.1)} \times 100\%$
	Data Transmission	Downlink Forwarding Rate on the TRAN Platform	$\frac{L.Traffic.SGW.Rx.SDU.Bytes}{VS.Gtpu.RxBytes} \times 100\%$ (The calculation takes all cells served by the base station into account.)
		Uplink Forwarding Rate on the TRAN Platform	$\frac{L.Traffic.SGW.Tx.SDU.Bytes}{VS.Gtpu.TxBytes} \times 100\%$ (The calculation takes all cells served by the base station into account.)
		Downlink PDCP Forwarding Rate of a Cell	$\frac{L.Thrp.bits.DL}{L.Traffic.SGW.Rx.SDU.Bytes} \times 8 \times 100\%$
		Uplink PDCP Forwarding Rate of a Cell	$\frac{L.Thrp.bits.UL}{L.Traffic.SGW.Tx.SDU.Bytes} \times 8 \times 100\%$
		ARQ DL Retrans Ratio	$\frac{L.Traffic.DL.AM.Retrans.TxPackets}{L.Traffic.DL.AM.TxPackets} \times 100\%$
		ARQ UL Retrans Ratio	$\frac{L.Traffic.UL.AM.Retrans.RxPackets}{L.Traffic.UL.AM.RxPackets} \times 100\%$

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
		Downlink MAC Retransmission Rate	$\frac{(L.Traffic.DL.SCH.QPSK.TB.Retrans + L.Traffic.DL.SCH.16QAM.TB.Retrans + L.Traffic.DL.SCH.64QAM.TB.Retrans)}{(L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB)} \times 100\%$
		Uplink MAC Retransmission Rate	$\frac{(L.Traffic.UL.SCH.QPSK.TB.Retrans + L.Traffic.UL.SCH.16QAM.TB.Retrans + L.Traffic.UL.SCH.64QAM.TB.Retrans)}{(L.Traffic.UL.SCH.QPSK.TB + L.Traffic.UL.SCH.16QAM.TB + L.Traffic.UL.SCH.64QAM.TB)} \times 100\%$
		PDCP Packet Discard Rate	$L.PDCP.Tx.Disc.Trf.SDU / L.PDCP.Tx.TotRev.Trf.SDU \times 100\%$
		PDCP Packet Loss Rate	$L.Traffic.DL.PktUuLoss.Loss / L.Traffic.DL.PktUuLoss.Tot \times 100\%$
		Downlink PDCP Delay(ms)	$L.Traffic.DL.PktDelay.Time / L.Traffic.DL.PktDelay.Num \times 100\%$
		DL Network Traffic Volume	$L.Thrp.bits.DL$
		UL Network Traffic Volume	$L.Thrp.bits.UL$
NSA	NSA Access	Success Rate of Outgoing Handovers for NSA PCC Anchoring	$L.NsaDC.PCCAnchor.HHO.ExecSuccOut / L.NsaDC.PCCAnchor.HHO.PrepAttOut \times 100\%$
		SgNB Addition Success Rate	$L.NsaDc.SgNB.Add.Succ / L.NsaDc.SgNB.Add.Att \times 100\%$
	NSA Service Drop	Abnormal SgNB Release Rate	$\frac{(L.NsaDc.SgNB.Rmv.Att - L.NsaDc.MenbTrig.SgNB.NormRel - L.NsaDc.SgnbTrig.SgNB.NormRel)}{L.NsaDc.SgNB.Rmv.Att} \times 100\%$
		SgNB Failure Rate	$L.NsaDc.ScgFailure / L.NsaDc.SgNB.Rmv.Att \times 100\%$

Service Type	Sub-Scenario	KPI Scenario	Calculation Formula
		Service Drop Rate of NSA UEs	$\frac{L.NsaDc.E-RAB.AbnormRel}{(L.NsaDc.E-RAB.AbnormRel + L.NsaDc.E-RAB.NormRel)} \times 100\%$
	NSA Handover	Success Rate of Outgoing Handovers for NSA UEs	$\frac{L.NsaDc.HHO.ExecSuccOut}{L.NsaDc.HHO.PrepAttOut} \times 100\%$
		NSA DC Pcell Change Success Rate	$\frac{L.NsaDc.PCell.Change.Succ}{L.NsaDc.PCell.Change.Exec} \times 100\%$

Users can log in to the FMA to view and download diagnosis results and fault information.

NOTE

- Automatic fault detection is not supported in NB-IoT or eMTC scenarios.
- Only KPI-based detection analysis is supported. Analysis on cell faults or base station disconnections is not supported.
- Network-wide automatic detection in all scenarios can be used to analyze a maximum of 4000 equivalent NEs.

Cell Unavailable Automatic Detection

The FMA supports automatic detection of cell faults. After a fault is detected, the FMA automatically performs fault diagnosis and information collection.

After automatic fault detection is enabled, the FMA automatically obtains the total number of cells with alarms reported within a specified measurement period and generates a report. If the number of alarms exceeds a threshold, it is regarded that a cell fault occurs. The threshold can be configured on the FMA by setting **Cell Unavailable Automatic Detection Conditions** under **Wireless Settings > Fault Auto Detection**.

The information collection association switch is turned on by default. After the switch is turned on, reports of automatic detection are automatically packaged for the information collection function.

The automatic detection information collection switch is turned on by default. After the switch is turned on, automatic fault detection tasks collect fault information about faulty NEs.

The FMA automatically performs fault diagnosis and information collection for faults detected.

- The FMA implements fault diagnosis on selected NEs and provides a diagnosis report. The report contains information about the faulty NEs, cell device, and cells with regular faults.

- For details about the data collected by the information collection function triggered by automatic detection, see [3.1.2.2.3 Information Collection](#).

Users can log in to the FMA to view and download diagnosis results and fault information.

3.1.2.2.7 Other Functions

Configuration Comparison

The analysis of a large number of base station faults shows that the configurations of faulty base stations or cells are different from those of normal base stations or cells. During troubleshooting and analysis, identifying key configuration differences can make fault rectification more efficient. Unlike traditional text comparison tools, the configuration comparison function can be used to compare parameter configurations of MOs between normal and abnormal base stations, helping users quickly identify the configuration differences that cause problems.

Subscribe Settings

This function allows you to subscribe to traffic statistics to ensure the proper implementation of functions that rely on NE configurations and traffic statistics.

If traffic statistics subscription is disabled, the subscription to related performance counters is cancelled and the FMA may fail to work properly.

If periodic traffic statistics subscription is disabled, new performance counters and performance counters of new NEs cannot be subscribed to by the FMA.

3.1.2.3 NR Wireless Fault Management

NR wireless fault management is implemented on the MAE to help users troubleshoot wireless faults on 5G networks.

NR wireless fault management obtains gNodeB configurations (the size of a configuration file does not exceed 200 MB), alarms, performance counters, and operation logs from the MAE and analyzes the fault. After an analysis is complete, the fault analysis results are provided to help users identify potential causes and accelerate service recovery. NR wireless fault management provides the following functions:

- **Fault overview**
Multisite faults are analyzed to help users locate the faulty base stations or cells. Users can continue to perform fast fault diagnosis or information collection.
- **Fast fault diagnosis**
After the faulty base stations or cells are determined using the fault overview function or manually selected, fault diagnosis can be performed on a single faulty base station or cell.
- **Information collection**
After the faulty base stations are determined using the fault overview function or manually selected, information about a single faulty base station can be collected.

- Key event assurance
Key event assurance monitors KPIs of multiple sites and displays related information on the dashboard.
- All scenario KPI overview
This function analyzes KPI trends of multiple sites in all scenarios and displays the results.
- Fault recovery confirmation
Fault recovery confirmation is supported in fast fault recovery scenarios.

Table 3-30 lists the supported NR wireless fault scenarios.

Table 3-30 NR wireless fault scenarios

Scenario	Sub-Scenario	Applicable Networking	Check Item
Performance KPI Fault	Access	SA	RRC Setup Success Rate
			NGSIG Connection Setup Success Rate
			QoS Flow Setup Success Rate
			Proportion of Normal NAS Processes
		NSA	SgNB Addition Success Rate
			SgNB Addition Number Decrease
	Service Drop	SA	Service Drop Rate
		NSA	SgNB Abnormal Release Rate
			SgNB-Triggered SgNB Abnormal Release Rate
	Data Transmission	SA, NSA	User Uplink Average Throughput (Mbit/s)
			Cell Uplink Average Throughput (Mbit/s)
			User Downlink Average Throughput (Mbit/s)
			Cell Downlink Average Throughput (Mbit/s)
	Handover	SA	Handover In Execution Success Rate
			Intra-Frequency Handover Out Execution Success Rate
			Intra-Frequency Handover Out Preparation Success Rate
			Inter-Frequency Handover Out Preparation Success Rate
			Handover In Preparation Success Rate

Scenario	Sub-Scenario	Applicable Networking	Check Item
		NSA	Inter-RAT Handover Out Success Rate (NR to LTE)
			Intra-SgNB PSCell Change Success Rate
			Inter-SgNB PSCell Change Success Rate
			SgNB PSCell Change Success Rate
	Interoperability	SA	Coverage-based NR2L Handover Execution Success Rate
			Coverage-based NR2L Handover Preparation Success Rate
			EPS Fallback Preparation Success Rate
			Handover-based EPS Fallback Execution Success Rate
			Handover-based EPS Fallback Preparation Success Rate
	VoNR	SA	Voice Call Drop Rate 5QI1
			Voice Call Drop Rate 5QI5
			Voice Service Access Success Rate 5QI1
			Voice Service Access Success Rate 5QI5
			Voice Service Handover Success Rate 5QI1
			Voice Service Handover Success Rate 5QI5
	Paging	SA	Proportion of Paging Messages Sent over the Uu Interface
			Number of Paging Messages Received over the NG Interface
			mt-Access RRC Setup Success Rate
Cell Fault	Cell Unavailable	SA, NSA	NR Cell Unavailable
	Cell Capability Degraded		NR DU Cell TRP Capability Degraded

Scenario	Sub-Scenario	Applicable Networking	Check Item
Fault Scenario	Fault Sub Scenario	SA, NSA	Clock Interference Fault
			Interference Fault

For details of how to use the FMA to handle NR wireless faults, see **Fault Management > FMA** in *iMaster MAE Product Documentation*.

 NOTE

When you group NEs in the topology view, it is recommended that the number of NEs in a group be less than 200. FMA fault analysis can be performed by group to improve efficiency.

3.1.2.3.1 Fault Overview

Fault overview provides an overview of overall faults to help users learn the entire network's operating status.

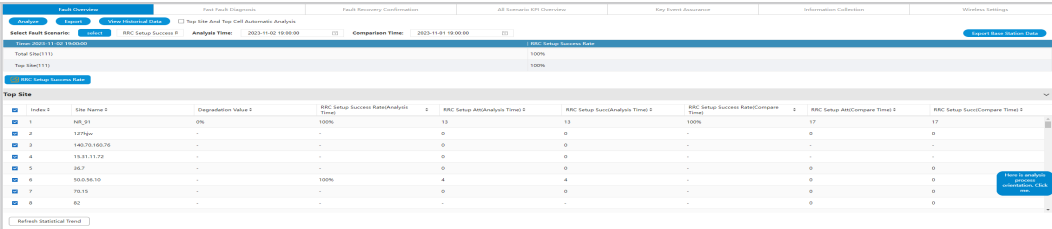
After users select fault scenarios (as listed in [Table 3-30](#)), NR sites to be analyzed, the fault occurrence time, and the comparison time, the fault overview function performs fault analysis. Fault overview provides different information for performance KPIs, cell faults, and fault scenarios.

Performance KPI Faults

For performance KPIs, the fault overview displays the following information.

- **Top sites**
Includes site name, degradation value, and the KPI values at the analysis time and comparison time, as shown in [Figure 3-23](#). A maximum of 200 top sites are supported.

Figure 3-23 Top site example



Index	Site Name	Degradation Value	RRC Setup Success Rate(Analysis Time)	RRC Setup Abort(Analysis Time)	RRC Setup SuccessRate(Compare Time)	RRC Setup Abort(Compare Time)
1	405_21	0%	100%	13	100%	17
2	147464	-	-	0	-	0
3	14676318076	-	-	0	-	-
4	15,91,11,79	-	-	0	-	-
5	86,7	-	-	0	-	0
6	1802-188-10	-	100%	4	-	0
7	79,15	-	-	0	-	0
8	82	-	-	-	-	0

- **KPI trend chart**
Includes the KPI changes on the fault occurrence day, the previous day, and the same day from the previous week.
- **KPI deterioration time**
Displayed on a KPI trend chart.
- **Failure cause value distribution and comparison with normal KPI time**

According to the KPI calculation formula, KPI failure cause values are decomposed and linked to related performance counters, and the KPIs at the fault occurrence time are compared with those at a specified time.

- Top sites that affect KPIs
For the sites that require further analysis, fast fault diagnosis or information collection can be performed, as shown in [Figure 3-24](#).

Figure 3-24 Example of top sites that affect KPIs

Site List		
NEs to Be Analyzed:10		
	Site Name	Contribution Rate
☒	59005G-32	23.88%
☒	59005G-6-12	23.23%
☒	59005G-6	23.23%
☒	5900-6	15.81%
☒	5900-32	13.85%
☒	wzh56	0%
☒	om88	0%
☒	OM95	0%
☒	HF32	0%
☒	BTS3900_LN_17	0%
☐	8169_y30065589	0%
☐	64.41	0%

 NOTE

- The FMA provides automatic analysis. If users select top sites and cells for automatic analysis and click **Analyze**, the FMA automatically performs fault overview and fast fault diagnosis. No manual intervention is required. If users do not select **Automatic Analysis** but click **Analyze**, the FMA only performs fault overview.
- "-" in the analysis result indicates that the corresponding counter data does not exist on the OSS.

Cell Faults

For cell faults, fault overview displays the following information:

- According to the selected faulty scenario, the numbers of sites and cells where the fault occurs are displayed. The alarms are classified into active alarms and historical alarms. [Figure 3-25](#) shows an example.

Figure 3-25 Example of the number of sites and cells with faults

Item		Cell Count Unavailable	
		Current Alarm	History Alarm
Site Num(s)	1	0	0
Cell Num	2	0	0
Get Cell Unavailable			
Fault Cell List			
Current Alarm		History Alarm	
Alarm Index	Alarm Details	Site Num	Cell Num
> 1	Abnormal NR DU cell resource	1	1
> 2	The Ethernet port on link is faulty	1	1
> 3	Abnormal P1 interface(CRLL, TDD)	1	-
> 4	gNodeB Out of Service	1	-
Total records: 4			
Refresh Statistical Trend			

 NOTE

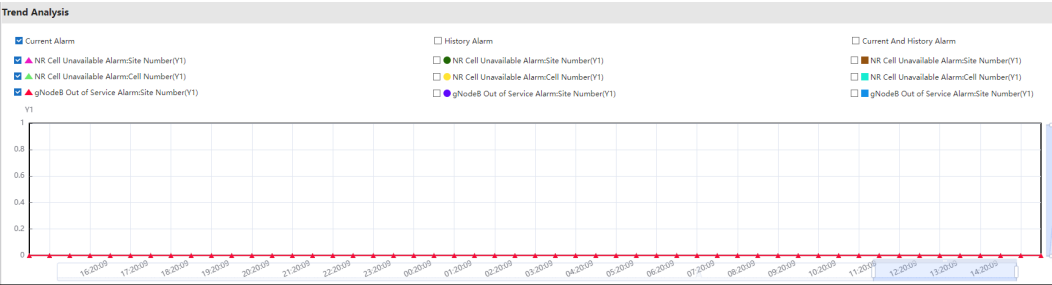
- Only historical alarms generated within 24 hours before the analysis are recorded.
- List of cells that report alarms: displays the number of sites and cells where an alarm related to the fault is reported. The alarms are classified into active alarms and historical alarms. [Figure 3-26](#) shows an example.

Figure 3-26 Example of the list of cells that report alarms



- Alarm quantity change trend: [Figure 3-27](#) shows an example.

Figure 3-27 Alarm quantity change trend example



Fault Scenarios

Fault scenarios involve clock interference faults and interference faults.

- Clock interference faults
For details about the design of the clock fault scenario in fault overview, see the design of the "NR Cell Unavailable" in the "Cell Fault" scenario. The faults are determined based on alarms. The following table lists the alarms to be checked. Only one alarm is needed to identify the "NR Cell Unavailable" fault while multiple alarms are required for identifying clock faults.

Table 3-31 Alarms used for clock fault analysis in fault overview

Alarm ID	Alarm Name
ALM-26261	External Clock Reference Not Configured
ALM-26262	External Clock Reference Problem
ALM-26124	GNSS Clock Output Unavailable
ALM-26125	GNSS Antenna Fault
ALM-26126	GNSS Locked Satellites Insufficient
ALM-26127	GNSS Maintenance Link Failure
EVT-26361	Forcible Synchronization with Reference Clock Phase
ALM-29830	gNodeB Out of Clock Synchronization

 NOTE

- By default, alarms on the entire network instead of only those of the selected sites are collected and analyzed. In addition, the alarms are collected at the site level but not the cell level.
- Analysis can be displayed in the form of trend chart based on historical or active alarms. You can also view the trend of a single alarm through alarm filtering.
- Interference faults

For interference faults, fault overview displays the following information:

- List of top interfered sites

The interference contribution rate of each gNodeB in the current NE group is displayed in descending order in the performance period starting from T_0 . Interference contribution rate = Difference between the interference of a site and the interference baseline value/Sum of the differences between the interference of all sites and the interference baseline value x 100%

- Top sites

By default, the number of top sites is 200. The NE name, degradation value, interference statistics, and minimum interference and noise received per uplink PRB of the top 200 sites are displayed.

 NOTE

Top N sites are filtered based on the deterioration of interference-related counter values and the deterioration of service KPIs of the corresponding NEs.

- Interference trend

The trend curve of the interference-related counter values of the top interfered NEs and the trend curve of the start time and end time of yesterday and last week are displayed.

- Export base station information

The exported base station information related to interference faults includes the NE name, interference statistics, minimum interference and noise per uplink PRB, and degradation value. Only the top 200 sites are displayed based on the KPI values.

3.1.2.3.2 Fast Fault Diagnosis

Fast fault diagnosis analyzes the performance of top sites identified using the fault overview function. The analysis result provides associated information about KPIs, active alarms, and operation logs for each of the top sites and each cell served by the top sites at the time when KPIs deteriorate. By analyzing KPIs, active alarms, and operation logs, this function helps users identify the minimum faulty units.

Fast fault diagnosis displays different diagnosis information for performance KPIs, cell faults, and fault scenarios.

For the detailed procedure of fast fault diagnosis, see "Starting Fast Fault Diagnosis" in *iMaster MAE Product Documentation*.

Performance KPI Faults

For performance KPI faults, the fast fault diagnosis function provides site- and cell-level diagnosis information, as described in [Table 3-32](#) and [Table 3-33](#).

For interference faults, the fast fault diagnosis function only provides cell-level diagnosis information.

Table 3-32 Site-level diagnosis information for performance KPI faults

Function	Description
Basic Site Information Table	• Site Name: Indicates the name of a site. • gNodeB CU X2 Interface ID: Indicates the ID of a gNodeB CU X2 interface in NSA networking, when the selected fault scenario is related to data transmission. • X2 Connection Status: Indicates the connectivity status of a peer LTE NE in NSA networking, when the selected fault scenario is related to data transmission. • Top anchor site ID (Failure Rate): Displays the gNodeB IDs and PSCell change failure rates of top sites with PSCell change failures in NSA networking, when the selected fault scenario is Inter-SgNB PSCell Change Success Rate or a KPI fault scenario related to access and service drop. • Top anchor site PLMN: Displays the PLMN IDs of top sites with PSCell change failures in NSA networking, when the selected fault scenario is Inter-SgNB PSCell Change Success Rate or a KPI fault scenario related to access and service drop. • Operator ID: Displays the operator IDs of top sites with PSCell change failures in NSA networking, when the selected fault scenario is Inter-SgNB PSCell Change Success Rate or a KPI fault scenario related to access and service drop. • Contribution rate: Indicates the ratio of failures on a site to the total failures, calculated when the fast fault diagnosis function analyzes KPIs. • License Analysis: Displays active license-related alarms at the fault occurrence time (only for non-data-transmission KPIs). • Version Change Info: Lists the latest software version changes of a site. • Risky Operation: Displays risky operation records in the specified period.

Function	Description
Transmission Diagnosis Info	Transmission diagnosis information is provided on the Original Messages and Diagnosis Results tab pages. After the completion of a diagnosis, the following information will be provided: • Transmission port information (the DSP ETHPORT command output) • ARP information learned by the transmission ports (the DSP ARP command output) • Configured route information (the output of the following commands) – DSP IPRT when the old transmission configuration model is used – DSP IPROUTE4 when the new transmission configuration model is used • Transmission quality of the Xn and NG interfaces (output of the PING command) • NG interface information (the DSP GNBCUNGINTERFACE command output) The preceding functions apply to the analysis of NGSIG Connection Setup Success Rate and Handover In Execution Success Rate. MML data in the transmission diagnosis information can be exported in CSV format. Users can click Export to export all MML data on the Diagnosis Results tab page.
Active Alarms	Displays the current active alarms of the to-be-analyzed site. NOTE Known alarms on specific sites can be used to identify other sites with the same alarms.
Fault Tree	Displays the preliminary fault analysis result for the current site and provides recovery suggestions.

Table 3-33 Cell-level diagnosis information for performance KPI faults

Function	Description
Basic Cell Information Table	• NrCellId indicates the ID of an NR cell (for non-data-transmission KPIs). • NrDuCellId indicates the ID of an NR DU cell (for data transmission KPIs). • Contribution Rate indicates the ratio of failures in a cell to the total failures, which is analyzed by the fast fault diagnosis function at a specified time. • Actual value of a KPI indicates the cell KPI value at the time when the KPI is analyzed. In the case of interference faults, the basic cell information table provides the values of current analyzed KPIs and associated service KPIs (average uplink PUSCH MCS of the cell) at the time when the analysis is performed. • Denominator (number of attempts) of a KPI indicates the number of setup attempts related to this KPI, if a non-traffic-volume-related KPI (such as success rate or service drop rate) is analyzed. • Heavily Loaded indicates whether a cell is heavily loaded based on CPU usage of the NE board serving the cell.
Cell Dashboard	Provides correlation analysis on the cell KPIs, alarms, and operation logs. After users click a time point when a KPI changes, the alarms and operation logs generated at the time will be displayed to help users quickly find the potential fault cause. Users can select records generated in the latest two hours before the fault occurs from operation logs for MAE-Deployment analysis. For cell faults, select records generated at the current time. NOTE The MAE-Deployment can only analyze records generated within two hours before the fault occurrence time, and can only analyze MOD, RMV, and ADD commands.
Board Diagnosis Info	Displays the CPU usage trends of the main control board and BBP serving an NR cell. This helps users observe the relationship between the cell KPI changes and the board CPU usage.
Cell KPI Analysis	Displays the trends of cell-level performance counters in an NR cell.

Cell Faults

For cell faults, the fast fault diagnosis function provides only site-level diagnosis information, as described in [Table 3-34](#).

Table 3-34 Diagnosis information for cell faults

Function	Description
Site Key Configuration Table	Lists information about each faulty cell served by the site, including: • Basic configuration: duplex mode, TX/RX mode, subframe ratio, and others • Operator information: tracking area ID, tracking area code, operator index, mobile country code, mobile network code, and others
Cell Device Info	Includes the numbers of sector devices, channels, baseband processing boards, and carriers on each TX channel.
Cell Dashboard	Lists the following information about a faulty cell: • Counters related to cell unavailability • Cell alarms • Cell operation logs (excluding those related to DSP and LST MML commands)
Fault Tree	Displays the preliminary fault analysis result on the analyzed cells and provides recovery suggestions.

Fault Scenarios

- Clock Interference Fault
Table 3-35 lists the results of fast fault diagnosis of clock interference faults.

Table 3-35 Diagnosis information in clock fault scenarios

Function	Description
List of sites with active alarms	Sites with active clock-related alarms on the entire network are displayed. (Table 3-31 lists the alarm list.)

Function	Description
Check the configuration of the selected analysis site	The FMA obtains the following information about the sites in the preceding site list through NR configuration or real-time MML query: • Basic information including the base station ID, NR DU cell ID, frame offset, duplex mode, downlink bandwidth, frequency band, and NR-ARFCN • Clock information including the clock source, clock synchronization mode, standard for clock mutual lock, setting of Clock-Unavailable Cell Activation Switch, setting of Clock Out-of-sync Detect Switch, slot configuration, and slot structure
KPI Analysis of Selected Sites	The following analysis charts for cells in the configuration check result are provided: • Symbol-level average receive energy curves of all selected cells • Uplink average interference curves of all selected cells • Symbol-level receive energy curve of the selected cell • Uplink interference curve of the selected cell

- Interference Fault

For interference faults, the fast fault diagnosis page provides only the **Cell** tab page and the diagnosis information is as described in [Table 3-36](#).

Table 3-36 Diagnosis information in interference fault scenarios

Function	Description
Key cell information table	Includes NR DU cell ID, degradation value, interference-related traffic statistics, and PUSCH MCS.
Cell dashboard	Provides associated analysis of NR cell interference statistics, PUSCH MCS, NR cell alarms, and operation logs.

ToB Faults

- ToB service exception

For ToB faults, the fast fault diagnosis function provides site- and cell-level diagnosis information, as described in [Table 3-37](#) and [Table 3-38](#).

Table 3-37 Site-level diagnosis information for ToB faults

Function	Description
Basic Site Information Table	• Site Name: Indicates the name of a site. • Version Change Info: Lists the latest software version changes of a site. • License Analysis: Displays active license-related alarms at the fault occurrence time. • Risky Operation: Displays risky operation records in the specified period.
Key MML Information	Key MML information is provided in the MML Information and Original Messages sheets. The query results of the following MML commands are displayed: • DSP GNBUEBASICINFO: UETYPE=ALL_UE; //Outputs information such as basic UE information, DRB list, cell list, and CA information. • DSP GNBUEBASICINFO: UETYPE=FAULTY_UE; • DSP GNBNETWORKSLICE; MML data in the key MML information can be exported in CSV format. Users can click Export to export all MML data on the MML Information tab page.
Active Alarms	Displays the active alarms of the to-be-analyzed site. NOTE Known alarms on specific sites can be used to identify other sites with the same alarms.
Parameter Check	Identifies core parameters related to ToB low-latency services of a specific site, and compares the parameter settings of the current site and baseline parameter settings.

Table 3-38 Cell-level diagnosis information for ToB faults

Function	Description
Basic Cell Information Table	• NrCellId: Indicates the ID of an NR cell. • Heavily Loaded: Indicates whether a cell is heavily loaded based on CPU usage of the NE board serving the cell.

Function	Description
Cell Dashboard	Provides correlation analysis on the cell KPIs, alarms, and operation logs. After users click a time point when a KPI changes, the alarms and operation logs generated at the time will be displayed to help users quickly find the potential fault cause. Users can select records generated in the latest two hours before the fault occurs from operation logs for MAE-Deployment analysis. For cell faults, select records generated at the current time. NOTE The MAE-Deployment can only analyze records generated within two hours before the fault occurrence time, and can only analyze MOD, RMV, and ADD commands.
Cell KPI Analysis	Displays the trends of cell-level, slice-level, 5QI-related, and RedCap-related indicators of an NR cell.

Descriptions of Fault Tree and Diagnosis Report

The fault tree displays the overall diagnosis results, as shown in [Figure 3-28](#). Diagnosis items highlighted in red indicate faults which can be linked to detailed diagnosis results in the sub-scenario. Diagnosis items in green indicate that no faults occur and no detailed results are provided.

For 5G NEs, the fault diagnosis function displays a fault tree for typical fault scenarios. This function implements tool-based associated data collection and analysis and supports one-click diagnosis and analysis of typical fault scenarios, facilitating fast fault recovery on the live network. The fault scenarios supported in this version include those related to access, service drops, data transmission, and cell unavailability, as described in [Table 3-39](#).

Figure 3-28 Fault tree example

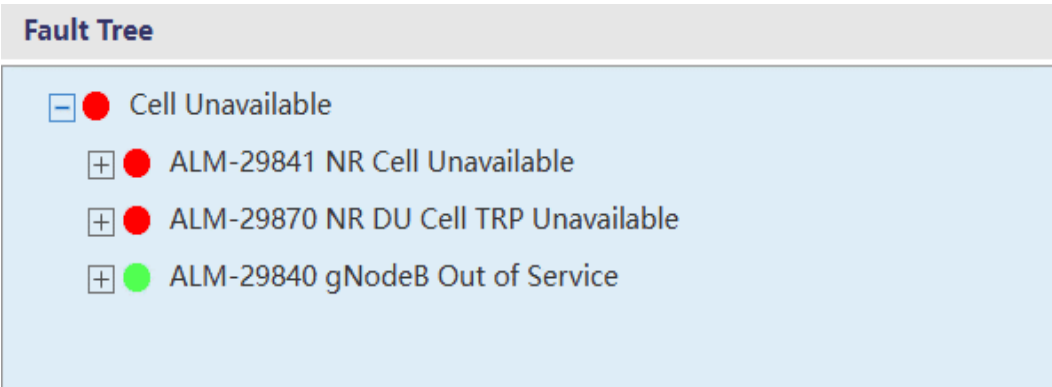


Table 3-39 Typical scenarios supported by the fault tree

First-Level Scenario	Second-Level Scenario	Third-Level Scenario
Access	eNodeB not initiating SgNB addition (NSA)	eNodeB not delivering 5G measurement configurations to the UE after UE accessing LTE network
		UE not sending B1 measurement reports
		B1 measurement reports related to NR not received by eNodeB
		eNodeB not initiating SCG addition after receiving B1 measurement reports related to NR
	SgNB addition rejected by gNodeB (NSA)	No available radio resources in NR cells
		Transmission resource unavailable
		NR cell unavailable
	Cause of QoS flow setup failure (SA)	TNL cause QosFlow setup fault
		Core network issues
		RNL issues
	E-RAB modification failure (NSA)	S1AP_ERAB_MOD_CONF not received
	The number of SCG additions decreases (NSA)	The number of SgNB additions accepted by NR decreases
	UE failing in random access on NR networks (NSA)	Incorrect NSA DC UE SCG UL Power Rate configuration on eNodeB
	The RRC connection setup fails (SA)	The RRC connection setup fails due to a channel calibration failure
		The RRC connection setup fails due to a large frequency offset of the clock
		Msg3 or msg5 fails due to strong interference
		Clock Out Of Step
		RRC connection setup request messages are discarded due to flow control

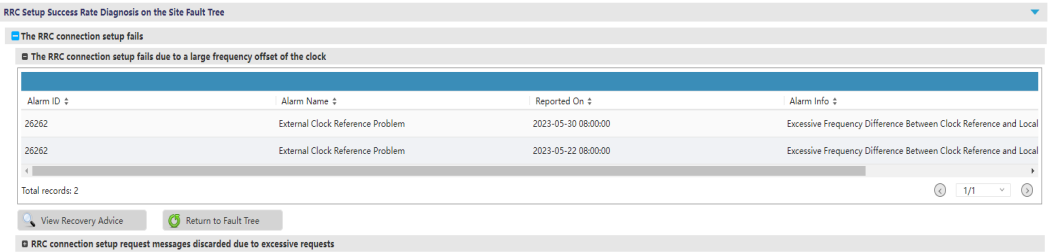
First-Level Scenario	Second-Level Scenario	Third-Level Scenario
		RRC connection setup request messages discarded due to excessive requests
		Failed to allocate baseband user resources
		Failed to allocate PUCCH resources
		The UE cannot receive the RAR as there is no downlink transmit power
	The NGSig setup fails (SA)	RelativeAMFCapacity Of Corenetwork Is Zero
		Insufficient Number Of Users License
		Ng trans fault
		Core network other issues
Data transmission	The user rate decreases	Signaling plane abnormal
		Air interface abnormal
		Base station abnormal
		The core network is abnormal
		Signaling plane abnormal
		Caused by abnormal alarm of transmission link
Service Drop	Service drops due to 5G-initiated abnormal releases in NSA networking	Service drops due to transmission exceptions
		Service drops caused by TNL issues
	Service drops caused by 5G-initiated abnormal releases in SA networking	Abnormal service drops of QoS flows caused by radio issues
		Abnormal service drops of QoS flows caused by TNL issues
	Service drops caused by TNL issues (NSA)	Service drops caused by S1-U transmission link faults
		Service drops caused by X2-U transmission link faults
		Service drops caused by the sending of a GTPU Err Ind message by the faulty core network

First-Level Scenario	Second-Level Scenario	Third-Level Scenario
	Abnormal service drops of QoS flows caused by TNL issues (SA)	Service drops caused by the sending of a GTPU Err Ind message by the faulty 4G anchor site
		Abnormal service drops of QoS flows caused by gNodeB NG-U transmission link faults
		Abnormal service drops of QoS flows caused by the sending of a GTPU Err Ind message by the faulty core network
		Abnormal service drops caused by external interference
	Abnormal service drops of QoS flows caused by radio issues (SA)	Abnormal service drops possibly caused by clock out-of-synchronization
Handover failure in SA networking	NR cell handover not triggered	Neighboring NR cell not configured on the NR side
		Neighboring LTE cell not configured on the NR side
Handover Failure in NSA Networking	NR Cell Handover not Triggered	Neighboring NR Cell PCI Conflict
		No A3 MR Control Sent To UE From LTE UU Interface
		No A3 MR Event Report From UE
	Inter-gNodeB Handover Failure	NR Inter-Site HO SgNB Change Required Reject By LTE
	Ue Fail To Access Target Nr Cell	ENodeB Received Msg3 Message But Does Not Send The Msg4(UL Grant) Message. As A Result, "Receive Content Timeout" Is Displayed In The KeyEvent Of The UE
	LTE Inter-Site HO Cause NR Release	LTE Inter-Site HO And Release NR Abnormally
Interoperability	Handover preparation success rate for EPS fallback decreases (SA)	Neighboring LTE cell configuration is incorrect
		The core network is abnormal
		The number of EPS fallback attempts decreases

First-Level Scenario	Second-Level Scenario	Third-Level Scenario
VoNR SA	Voice Service Access Success Rate 5QI1	The base station is abnormal
	Voice Service Access Success Rate 5QI5	The base station is abnormal
Cell Fault NOTE As there are a large number of third-level scenarios under this scenario, this table lists only some typical ones.	ALM-29841 NR Cell Unavailable	DU cell is unavailable
		Root Alarms
		F1 fault
	ALM-29870 NR DU Cell TRP Unavailable	The power configuration is incorrect
		The bandwidth configuration is incorrect
		The frequency configuration is incorrect
		The RF unit is abnormal
		Incorrect Sector Equipment Configurations
		The baseband unit is abnormal
		Insufficient CPRI bandwidth
		Insufficient IDX2 bandwidth
		Configuration conflict
		The license resources are insufficient
	ALM-29840 gNodeB Out of Service	ALM-26261 External Clock Reference Not Configured
		ALM-26262 External Clock Reference Problem
		ALM-29816 gNodeB NG Control Plane Transmission Interruption
		ALM-29815 gNodeB NG Interface Fault
	NR DU Cell TRP Capability Degraded	Antenna channel exceptions
		ALM-26235 RF Unit Maintenance Link Failure
		Insufficient license

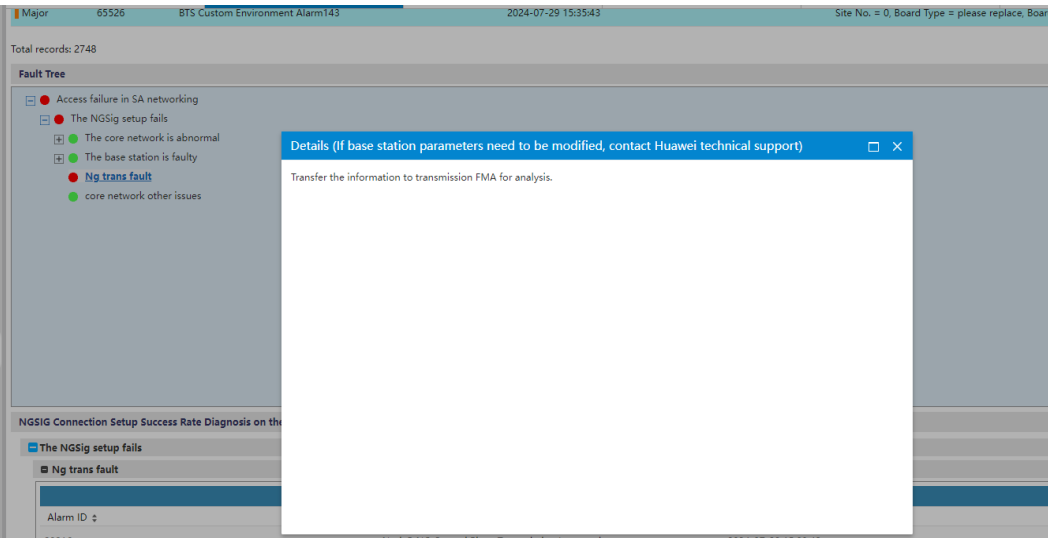
Figure 3-29 shows detailed diagnosis results in a sub-scenario.

Figure 3-29 Detailed diagnosis result example in a sub-scenario



Corresponding recovery advice is provided as a reference after fault diagnosis in a sub-scenario. For details, see [Figure 3-30](#).

Figure 3-30 Recovery advice example in a sub-scenario



3.1.2.3.3 Information Collection

This function can quickly and accurately collect onsite information and improve fault recovery efficiency.

- An information collection task involves the NR, RAN sharing, 5GtoB, 4G-relation, and AutoDetect scenarios. By default, fault information in NR scenarios is preferentially collected and that in other scenarios is collected in sequence. To preferentially collect information in other scenarios, select **NR first**, **RanSharing first**, or **5GtoB first** when creating a task.
- If the site is an NSA site, the NIC collects the first-batch data on the LTE side by default.
- Related information and operations are provided in the following columns for a created fault information collection task: **Task Name**, **Progress**, **Status**, **Sources**, **Execution Time**, **Run Time**, **Created On**, **Created By**, and **Operation**. The value of **Sources** can be **Manual** or **Auto Detect**. For manually created tasks, the **Details**, **Stop**, and **Delete** buttons in the **Operation** column can be used to query task details, stop the task, and delete the task, respectively. A stopped task can be deleted but cannot be resumed. For automatic detection tasks, only the **Download** button is available in the **Operation** column.

- The automatic detection results collected by the NIC can be displayed.

Table 3-40 lists the collected fault information.

Table 3-40 Collected fault information

Collection Item	Scope
NE configuration file	One file per NE
MML query output	One file per NE
Results of pinging a base station from the MAE	One file per NE
Results of tracing routes from the MAE to a base station	One file per NE
Performance indicators	During a specified collection period
Alarm data	During a specified collection period
MAE operation logs	During a specified collection period
Original operation logs of top 5 NEs	One file per NE
External CHR subscription records	One file for multiple NEs (a maximum of five NEs)
Automatic detection report (only involved in manually created information collection tasks)	Consisting of reports of Network-wide Automatic Detection in All Scenarios and Cell Unavailable Automatic Detection within seven days (A maximum of 100 MB reports can be collected.)

In NSA scenarios, when users select an NR NE, NR wireless fault management automatically identifies the peer anchor eNodeB and collects associated fault information. The collection items for the anchor eNodeB are the same as that for the NR NE, as listed in **Table 3-40**.

For details, see "Network Information Collection" in *iMaster MAE Product Documentation*.

3.1.2.3.4 All Scenario KPI Overview

This function supports site-level KPI and performance counter analysis for NEs managed by the MAE. The supported scenarios are **NSA Signaling, SA Intra-RAT Signaling, SA Inter-RAT Signaling, Voice, Rate & Interference, RedCap**, and **MOCN**. Specifically,

- Users can select NEs from the network topology or select a version group on the **All Scenario KPI Overview** tab page for analysis. When the KPI period is set to **P60**, a maximum of 2500 NEs can be analyzed at a time. For other KPI periods, the maximum number of NEs that can be analyzed at a time is displayed on the GUI. It is recommended that no more than one user use the

all scenario KPI overview function at the same time. In addition, when using this function, you are not advised to use other fault analysis functions.

- Users can select the analysis date (the current day is selected by default). The analysis period is [00:00, 24:00), which is fixed to 24 hours. After the analysis is complete, the KPI statistics of the current day (corresponding to **today**), the previous day (corresponding to **yesterday**), and the same day of the previous week (corresponding to **last week**) are displayed by default for comparison.
- Users can specify a KPI period to view the statistical data in different KPI periods. The supported values on the GUI are **P5**, **P15**, **P30**, and **P60**, which correspond to 5 minutes, 15 minutes, 30 minutes, and 60 minutes, respectively. The default value is **P60**. Users need to ensure that the selected KPI period has been subscribed to.
- Users can export the analysis results. If the selected NEs are changed, the data analyzed in the current session of all scenario KPI overview can be exported using fault overview, fast fault diagnosis, fault recovery confirmation, and key event assurance.

Table 3-41 KPIs or performance counters applicable to NR

Scenario	KPI/Performance Counter
NSA Signaling	SgNB Addition Success Rate(%)
	Inter-SgNB PSCell Change Success Rate(%)
	Intra-SgNB PSCell Change Success Rate(%)
	SgNB-triggered Abnormal SgNB Released Rate(Only gNB Triggered) (%)
	SgNB Abnormal Release Rate(Including eNB&gNB Triggered) (%)
SA Intra-RAT Signaling	RRC Setup Success Rate(SA)(%)
	NgSig Setup Success Rate(SA)(%)
	QosFlow Setup Success Rate(SA)(%)
	Intra-Frequency Handover Out Execution Success Rate(%)
	Intra-Frequency Handover Out Preparation Success Rate(%)
	Inter-Frequency Handover Out Execution Success Rate(%)
	Inter-Frequency Handover Out Preparation Success Rate(%)

Scenario	KPI/Performance Counter
	Handover In Execution Success Rate(%)
	Handover In Preparation Success Rate(%)
	Service Call Drop Rate(%)
	Service Call Drop Rate with AMF(%)
SA Inter-RAT Signaling	EPS Fallback Preparation Success Rate(%)
	Handover-based EPS Fallback Preparation Success Rate(%)
	Handover-based EPS Fallback Execution Success Rate(%)
	Coverage-based NR2L HO Preparation Success Rate(%)
	Coverage-based NR2L HO Execution Success Rate(%)
Voice	Voice Service Access Success Rate 5QI1(%)
	Voice Service Access Success Rate 5QI5(%)
	N.Paging.Dis.PchCong
	Voice Call Drop Rate 5QI1(%)
	Voice Call Drop Rate 5QI5(%)
	Voice Service Handover Success Rate 5QI1(%)
	Voice Service Handover Success Rate 5QI5(%)
Rate & Interference	Cell Downlink Average Throughput(Mbps)
	Cell Uplink Average Throughput(Mbps)
	User Downlink Average Throughput(Mbps)
	N.ThpVol.DL(Kbit)
	User Uplink Average Throughput(Mbps)

Scenario	KPI/Performance Counter
	N.ThpVol.UL(Kbit)
	N.UL.NI.Avg(dBm)
	N.UL.Last.Symbol13.Pwr(dBm)
	N.UL.Last.Symbol6.Pwr(dBm)
	N.GAP.LastSymbol.Pwr(dBm)
	VS.Gtpu.RxDropPkts
RedCap	RedCap RRC Setup Success Rate(%)
	RedCap NgSig Setup Success Rate(%)
	RedCap QosFlow Setup Success Rate(%)
	RedCap Service Drop Rate(%)
	RedCap Intra Freq HO Out Success Rate(%)
	RedCap Inter Freq HO Out Success Rate(%)
	RedCap Average User Number
	RedCap NR2L HO Out Success Rate(%)
	Downlink Traffic Volume of PDCP Services of RedCap Ues(GB)
	Uplink Traffic Volume of PDCP Services of RedCap Ues(GB)
	RedCap User DL Experienced Rate(Excluding Last Slot Packets) (Mbit/s)
	RedCap User UL Experienced Rate(Excluding Small Packets)(Mbit/s)
	Average Downlink PRB Rate of RedCap BWP(%)
	Average Uplink PRB Rate of RedCap BWP(%)
	Average Downlink PRB Rate of RedCap Ues(%)
	Average Uplink PRB Rate of RedCap Ues(%)

Scenario	KPI/Performance Counter
MOCN	SgNB Addition Success Rate PLMN(%)
	SgNB-triggered Abnormal SgNB Released Rate PLMN(%)
	Abnormal SgNB Released Rate(Include MeNB triggered) PLMN(%)
	SA NgSig Setup Success Rate PLMN(%)
	QoS Flow Setup Success Ratio PLMN(%)
	VONR QoS Flow Setup Success Ratio PLMN(%)
	SA QoSFlow Drop Rate PLMN(%)
	VONR SA QoSFlow Drop Rate PLMN(%)
	DL RLC Traffic PLMN(GB)
	UL RLC Traffic PLMN(GB)
	N.User.RRCCConn.Avg.Operator

3.1.2.3.5 Key Event Assurance

This function monitors KPIs and displays related information on the dashboard, helping users obtain network running information in real time, such as license information, board temperature, alarms, and KPIs.

- NE specifications: A maximum of 100 sites, each supporting 10 cells, are supported. The maximum time range is 12 hours. If more than 100 sites require key event assurance, this function is not recommended.
- KPI specifications: Currently, eight types of KPIs associated with 5- or 15-minute period counters are supported. Before using this function, you need to subscribe to short-period counters on the performance management module of the MAE-Access.
- Eight types of KPIs are as follows: average and maximum main control board CPU usages, average and maximum baseband processing unit CPU usages, number of UEs in a cell, cell throughput, access success rate, call drop rate, SgNB addition success rate, and SgNB call drop rate. You can sort KPI information based on the sequence in which the KPIs are selected. For example, information of the first selected KPI is listed at the top.
- Six lines in the KPI monitoring trend chart for each KPI: one line showing the overall KPI trend of all sites and five lines each showing the trend of a top site
- Display of top sites for a KPI selected in the lower part of the trend chart: You can select a site in the list to view the related dashboard which provides the

KPI trend chart, license information, board temperature, and alarm information.

- Switch between tab pages on the dashboard to monitor site-level and cell-level data for the following five types of KPIs: cell throughput, access success rate, call drop rate, SgNB addition success rate, and SgNB call drop rate.
- Data export in the html format: The layout and content of the output is the same as the displayed content.

3.1.2.3.6 Fault Recovery Confirmation

The FMA supports fault recovery confirmation in fast fault recovery scenarios. It showcases the changes in KPIs or cell status after the recovery plan is executed. The changes of NE KPIs, cell status, and alarm status are displayed in visualized graphics on the dashboard. Fault handling personnel can use such data to determine whether a fault is rectified. The following table lists the detailed functions.

Function	Sub-Function	Description
Fault recovery confirmation	1-minute KPI trend	Displays the 1-minute KPI trend after the recovery plan is executed, helping users to determine whether the fault is recovered.
	List of alarm clearance status	Displays the alarm clearance status after the recovery plan is executed, helping users to determine whether the fault is recovered.
	List of cell status	Displays the status of the cell instance after the recovery plan is executed, helping users to determine whether the fault is recovered.
	Saving of the recovery confirmation results	Saves the KPI monitoring results, cell fault alarm clearance results, and cell status results.

3.1.2.3.7 Other Functions

Automatic Fault Detection

The FMA supports automatic detection of cell faults and KPI faults for NR networks. After a fault is detected, the FMA automatically performs fault diagnosis and information collection.

Network-wide Automatic Detection in All Scenarios

The FMA supports periodic subscription to KPIs and automatic KPI analysis through the MAE-Access. When one or multiple associated KPIs on the network are abnormal and the number of abnormal sites reaches a certain value, the FMA can automatically identify and diagnose the exceptions, generating an analysis

report for quick review. The report provides KPI analysis, alarm-related fault analysis, analysis of impact scope, and analysis related to operation and configuration, helping users to quickly rectify faults. The network-wide automatic detection in all scenarios function is enabled by default.

The information collection association switch is turned on by default. After the switch is turned on, reports of automatic detection are automatically packaged for the information collection function.

The automatic detection information collection switch is turned on by default. After the switch is turned on, automatic fault detection tasks collect fault information about faulty NEs.

You can set the thresholds for different scenarios on the **Scenario threshold** tab page. [Table 3-42](#) lists the details.

Table 3-42 Parameters related to automatic fault detection

Parameter	Description	Setting Notes
FMA Fault Analysis Interval	Minimum interval for network-level fault analysis. The default value is 2 hours.	The smaller the value of this parameter, the more the network faults that can be determined within 24 hours. The larger the value of this parameter, the fewer the network faults that can be determined within 24 hours. The default value is recommended.

Parameter	Description	Setting Notes
Level-1 Entry Threshold	Thresholds used to determine whether level-2 thresholds are required for further determination. If the KPIs of the current cell reach the level-1 entry thresholds, further analysis against level-2 thresholds is not required and analysis can be performed subsequently.	• For success rate-related KPIs, the larger the threshold values, the higher the probability of determining network faults. The smaller the threshold values, the lower the probability of determining network faults. • For deterioration rate-related KPIs, the larger the threshold values, the lower the probability of determining network faults. The smaller the threshold values, the higher the probability of determining network faults.

Parameter	Description	Setting Notes
Level-1 Exit Threshold	Thresholds used to determine whether the KPIs of the current cell are normal. If the exit thresholds are reached, the KPIs of the current cell are normal and no further analysis is required.	- For success rate-related KPIs, the larger the threshold values, the more the KPIs being judged against the level-2 thresholds. The smaller the threshold values, the fewer the KPIs being judged against the level-2 thresholds. - For deterioration rate-related KPIs, the larger the threshold values, the fewer the KPIs being judged against the level-2 thresholds. The smaller the threshold values, the more the KPIs being judged against the level-2 thresholds. - The more the KPIs being judged against the level-2 thresholds, the lower the analysis efficiency, and the higher the analysis accuracy.
Minimum Sample Size for Level-1 Entry Threshold	Minimum numbers of samples used for determination against the level-1 entry thresholds, which prevents misjudgment caused by KPI value fluctuations when the number of UEs is small.	The larger the numbers, the stronger the anti-fluctuation capability. The smaller the numbers, the weaker the anti-fluctuation capability. The default values are recommended.

Parameter	Description	Setting Notes
Level-2 z-score Entry Threshold	Level-2 z-score thresholds determined based on the statistics in the historical periods. If the KPI changes of the cell in the current period exceed the thresholds, the system determines that a fault occurs and exits the check.	The larger the absolute values of the thresholds, the higher the tolerance to KPI value fluctuation. The smaller the absolute values of the thresholds, the lower the tolerance to KPI value fluctuation. The default values are recommended.
Minimum Number of Cells to Determine a Network Fault	Minimum numbers of cells for determining a network fault.	The larger the numbers, the lower the probability of determining a network fault. The smaller the numbers, the higher the probability of determining a network fault. The default values are recommended.
z-score Entry Threshold for a Network Fault	z-score thresholds for the numbers of cells. If the number of cells with a specific KPI fault reaches the corresponding threshold, the fault is considered as a network-level fault.	The larger the absolute values of the thresholds, the higher the tolerance to KPI value fluctuation. The smaller the absolute values of the thresholds, the lower the tolerance to KPI value fluctuation. The default values are recommended.

 NOTE

As adjusting the preceding thresholds may cause missing or excessive detection of network faults, adjust the thresholds under the guidance of Huawei engineers.

The FMA analyzes basic network KPIs every 60 minutes. If KPIs are abnormal for a large number of base stations, the FMA generates an auto-detection analysis report. [Table 3-43](#) lists the KPI scenarios defined by the MAE-Access.

Table 3-43 KPI scenarios defined by the MAE-Access

Service Type	Sub-Scenario	KPI	Formula
Basic (SA)	Access	RRC Setup Success Rate	$\frac{N.RRC.SetupReq.Succ}{N.RRC.SetupReq.Att} \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		NGSIG Connection Setup Success Rate	$\frac{N.NGSig.ConnEst.Succ}{N.NGSig.ConnEst.Att} \times 100\%$
		QoS Flow Setup Success Rate	$\frac{N.QoSFlow.Est.Succ}{(N.QoSFlow.Est.Att - N.QoSFlow.Est.Att.EPSFB - N.QoSFlow.Est.Att.EmcFB)} \times 100\%$
		Ratio of Ngsig Setup	$\frac{N.NGSig.ConnEst.Att}{N.RRC.SetupReq.Att} \times 100\%$
		Number of Users	$N.User.RRCConn.Avg \times 100\%$
	Service Drop	Service Drop Rate	$\frac{N.QoSFlow.AbnormRel}{(N.QoSFlow.NormRel + N.QoSFlow.AbnormRel)} \times 100\%$
		UE Context Drop Rate	$\frac{N.UECntx.AbnormRel}{(N.UECntx.AbnormRel + N.UECntx.NormRel)} \times 100\%$
		PDU Session Drop Rate	$\frac{N.PDUSession.AbnormRel}{(N.PDUSession.AbnormRel + N.PDUSession.NormRel)} \times 100\%$
	Handover	Intra-Frequency Handover Out Execution Success Rate	$\frac{(N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut + N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut + N.HO.IntraFreq.IntragNB.ExecSuccOut)}{(N.HO.IntraFreq.Ng.IntergNB.ExecAttOut + N.HO.IntraFreq.IntragNB.ExecAttOut + N.HO.IntraFreq.Xn.IntergNB.ExecAttOut)} \times 100\%$
		Handover In Execution Success Rate	$\frac{(N.HO.Ng.IntergNB.ExecSuccIn + N.HO.IntragNB.ExecSuccIn + N.HO.Xn.IntergNB.ExecSuccIn)}{(N.HO.Ng.IntergNB.ExecAttIn + N.HO.IntragNB.ExecAttIn + N.HO.Xn.IntergNB.ExecAttIn)} \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		Inter-RAT Handover Out Success Rate (NR to LTE)	$\frac{N.HO.InterRAT.N2E.ExecSuccOut}{N.HO.InterRAT.N2E.ExecAttOut} \times 100\%$
		Intra-Frequency Handover Out Preparation Success Rate	$\frac{(N.HO.IntraFreq.Ng.IntergNB.ExecAttOut + N.HO.IntraFreq.Xn.IntergNB.ExecAttOut + N.HO.IntraFreq.IntragNB.ExecAttOut)}{(N.HO.IntraFreq.Ng.IntergNB.PrepAttOut + N.HO.IntraFreq.Xn.IntergNB.PrepAttOut + N.HO.IntraFreq.IntragNB.PrepAttOut)} \times 100\%$
		Inter-Frequency Handover Out Preparation Success Rate	$\frac{(N.HO.InterFreq.IntragNB.ExecAttOut + N.HO.InterFreq.Xn.IntergNB.ExecAttOut + N.HO.InterFreq.Ng.IntergNB.ExecAttOut)}{(N.HO.InterFreq.IntragNB.PrepAttOut + N.HO.InterFreq.Xn.IntergNB.PrepAttOut + N.HO.InterFreq.Ng.IntergNB.PrepAttOut)} \times 100\%$
		Handover In Preparation Success Rate	$\frac{(N.HO.Xn.IntergNB.ExecAttIn + N.HO.IntragNB.ExecAttIn + N.HO.Ng.IntergNB.ExecAttIn)}{(N.HO.Xn.IntergNB.PrepAttIn + N.HO.IntragNB.PrepAttIn + N.HO.Ng.IntergNB.PrepAttIn)} \times 100\%$
	Interoperability	EPS Fallback Preparation Success Rate	$\frac{N.VoiceFB.RespSucc}{N.Rcv.VoiceFB.Trig} \times 100\%$
		Handover-based EPS Fallback Preparation Success Rate	$\frac{N.HO.InterRAT.N2E.EPSFB.ExecAttOut}{N.HO.InterRAT.N2E.EPSFB.PrepAttOut} \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		Handover-based EPS Fallback Execution Success Rate	$\frac{N.HO.InterRAT.N2E.EPSFB.ExecSuccOut}{N.HO.InterRAT.N2E.EPSFB.ExecAttOut} \times 100\%$
		Coverage-based NR2L Handover Preparation Success Rate	$\frac{N.HO.InterRAT.N2E.Coverage.ExecAttOut}{N.HO.InterRAT.N2E.Coverage.PrepAttOut} \times 100\%$
		Coverage-based NR2L Handover Execution Success Rate	$\frac{N.HO.InterRAT.N2E.Coverage.ExecSuccOut}{N.HO.InterRAT.N2E.Coverage.ExecAttOut} \times 100\%$
	Data Transmission	User Uplink Average Throughput	$1000 \times \frac{(N.ThpVol.UL - N.ThpVol.UE.UL.SmallPkt)}{N.ThpTime.UE.UL.RmvSmallPkt}$
		Cell Uplink Average Throughput	$1000 \times \frac{N.ThpVol.UL.Cell}{N.ThpTime.UL.Cell}$
		User Downlink Average Throughput	$1000 \times \frac{(N.ThpVol.DL - N.ThpVol.DL.LastSlot)}{N.ThpTime.DL.RmvLastSlot}$
		Cell Downlink Average Throughput	$1000 \times \frac{N.ThpVol.DL.Cell}{N.ThpTime.DL.Cell}$
		Downlink Traffic Volume	$N.ThpVol.DL/1000/1000$
		Uplink Traffic Volume	$N.ThpVol.UL/1000/1000$
		PDCCP DL Packet Loss Rate	$\frac{(N.PDCP.DL.TrfSDU.TxPacket.Discard - N.PDCP.DL.TrfSDU.TxPacket.Discard.GTPU)}{N.PDCP.DL.TrfPDU.TxPackets} \times 100\%$
		PDCCP UL Packet UU Loss Rate	$\frac{N.PDCP.UL.TrfSDU.RxPacket.Loss}{(N.PDCP.UL.TrfPDU.RxPackets + N.PDCP.UL.TrfSDU.RxPacket.Loss)} \times 100\%$
	RedCap	RedCap QoS Flow Setup Success Rate	$\frac{N.QosFlow.Est.Succ.RedCap}{N.QosFlow.Est.Att.RedCap} \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		RedCap NGSIG Connection Setup Success Rate	$\frac{N.NGSig.ConnEst.Succ.RedCap}{N.NGSig.ConnEst.Att.RedCap} \times 100\%$
		RedCap RRC Setup Success Rate	$\frac{N.RRC.SetupReq.Succ.RedCap}{N.RRC.SetupReq.Att.RedCap} \times 100\%$
		RedCap Service Drop Rate	$\frac{N.QosFlow.AbnormRel.RedCap}{(N.QosFlow.AbnormRel.RedCap + N.QosFlow.NormRel.RedCap)} \times 100\%$
		RedCap Handover Out Success Rate (NR to LTE)	$\frac{N.HO.InterRAT.N2E.ExecSuccOut.RedCap}{N.HO.InterRAT.N2E.PrepAttOut.RedCap} \times 100\%$
		RedCap Inter Freq HO Out Success Rate	$\frac{(N.HO.InterFreq.IntragNB.ExecSuccOut.RedCap + N.HO.InterFreq.Xn.IntergNB.ExecSuccOut.RedCap + N.HO.InterFreq.Ng.IntergNB.ExecSuccOut.RedCap)}{(N.HO.InterFreq.IntragNB.PrepAttOut.RedCap + N.HO.InterFreq.Xn.IntergNB.PrepAttOut.RedCap + N.HO.InterFreq.Ng.IntergNB.PrepAttOut.RedCap)} \times 100\%$
		RedCap Intra Freq HO Out Success Rate	$\frac{(N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut.RedCap + N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut.RedCap + N.HO.IntraFreq.IntragNB.ExecSuccOut.RedCap)}{(N.HO.IntraFreq.IntragNB.PrepAttOut.RedCap + N.HO.IntraFreq.Xn.IntergNB.PrepAttOut.RedCap + N.HO.IntraFreq.Ng.IntergNB.PrepAttOut.RedCap)} \times 100\%$
		Downlink Traffic Volume of PDCP Services of RedCap UEs	$N.PDCP.Vol.DL.TrfSDU.Rx.RedCap/1000/1000/8$

Service Type	Sub-Scenario	KPI	Formula
		Uplink Traffic Volume of PDCP Services of RedCap UEs	$N.PDCP.Vol.UL.TrfSDU.Tx.RedCap/1000/1000/8$
		RedCap User DL Experienced Rate	$(N.ThpVol.DL.RedCap - N.ThpVol.DL.LastSlot.RedCap)/N.ThpTime.DL.RmvLastSlot.RedCap \times 1000$
		RedCap User UL Experienced Rate	$(N.ThpVol.UL.RedCap - N.ThpVol.UE.UL.SmallPkt.RedCap)/N.ThpTime.UE.UL.RmvSmallPkt.RedCap \times 1000$
NSA	NSA Access	SgNB Addition Success Rate	$N.NsaDc.SgNB.Add.Succ/N.NsaDc.SgNB.Add.Att \times 100\%$
		SgNB Addition Number Decrease	$N.NsaDc.SgNB.Add.Att$
		SgNB Addition Preparation Fail Rate	$(N.NsaDc.SgNB.Add.Att - N.NsaDc.SgNB.Add.Ack)/N.NsaDc.SgNB.Add.Att \times 100\%$
		SgNB Addition Execution Fail Rate	$(N.NsaDc.SgNB.Add.Ack - N.NsaDc.SgNB.Add.Succ)/N.NsaDc.SgNB.Add.Att \times 100\%$
	NSA Service Drop	SgNB-Triggered SgNB Abnormal Release Rate	$N.NsaDc.SgNB.AbnormRel/N.NsaDc.SgNB.Rel \times 100\%$
		SgNB Abnormal Release Rate	$(N.NsaDc.SgNB.Rel - N.NsaDc.SgNB.Rel.SgNBTrigger - N.NsaDc.SgNB.Rel.MeNBTrigger.NormalRel + N.NsaDc.SgNB.AbnormRel - N.NsaDc.SgNB.Rel.InitialAdd.RachFail)/N.NsaDc.SgNB.Rel \times 100\%$
	NSA Handover	Intra-SgNB PSCell Change Success Rate	$N.NsaDc.IntraSgNB.PSCell.Change.Succ/N.NsaDc.IntraSgNB.PSCell.Change.Att \times 100\%$

Service Type	Sub-Scenario	KPI	Formula
		Inter-SgNB PSCell Change Success Rate	$\frac{N.NsaDc.InterSgNB.PSCell.Change.Succ}{N.NsaDc.InterSgNB.PSCell.Change.Att} \times 100\%$
		InterSgNB Transmission Faults Ratio	$\frac{N.NsaDc.InterSgNB.PSCell.Change.Fail.Trans}{(N.NsaDc.InterSgNB.PSCell.Change.Att - N.NsaDc.InterSgNB.PSCell.Change.Succ)} \times 100\%$

The data collected in fault information collections triggered by automatic detections is consistent with that manually collected on the FMA. For details, see [3.1.2.3.3 Information Collection](#).

Cell Unavailable Automatic Detection

The FMA supports automatic detection of cell faults. After a fault is detected, the FMA automatically performs fault diagnosis and information collection.

After automatic fault detection is enabled, the FMA automatically obtains the total number of cells with alarms reported within a specified measurement period and generates a report. If the number of alarms exceeds a threshold, it is regarded that a cell fault occurs. The threshold can be configured on the FMA by setting **Cell Unavailable Automatic Detection Conditions** under **Wireless Settings > Fault Auto Detection > Cell Unavailable Automatic Detection Settings**.

The information collection association switch is turned on by default. After the switch is turned on, reports of automatic detection are automatically packaged for the information collection function.

The automatic detection information collection switch is turned on by default. After the switch is turned on, automatic fault detection tasks collect fault information about faulty NEs.

The FMA automatically performs fault diagnosis for faults detected to generate analysis reports as well as information collection for faulty NEs. For details about information collection, see [3.1.2.3.3 Information Collection](#).

Users can log in to the FMA to view and download diagnosis results and fault information.

Configuration Comparison

The analysis of a large number of base station faults shows that the configurations of faulty base stations or cells are different from those of normal base stations or cells. During troubleshooting and analysis, identifying key configuration differences can make fault rectification more efficient. Unlike traditional text comparison tools, the configuration comparison function can be

used to compare parameter configurations of MOs between normal and abnormal base stations, helping users quickly identify the configuration differences that cause problems.

Parameter Settings

The 5G online FMA supports customized settings of KPI thresholds and formulas corresponding to performance KPI fault sub-scenarios in fault overview.

Interference Identification Settings

The 5G online FMA supports customized settings of parameters related to interference fault scenarios in fault overview.

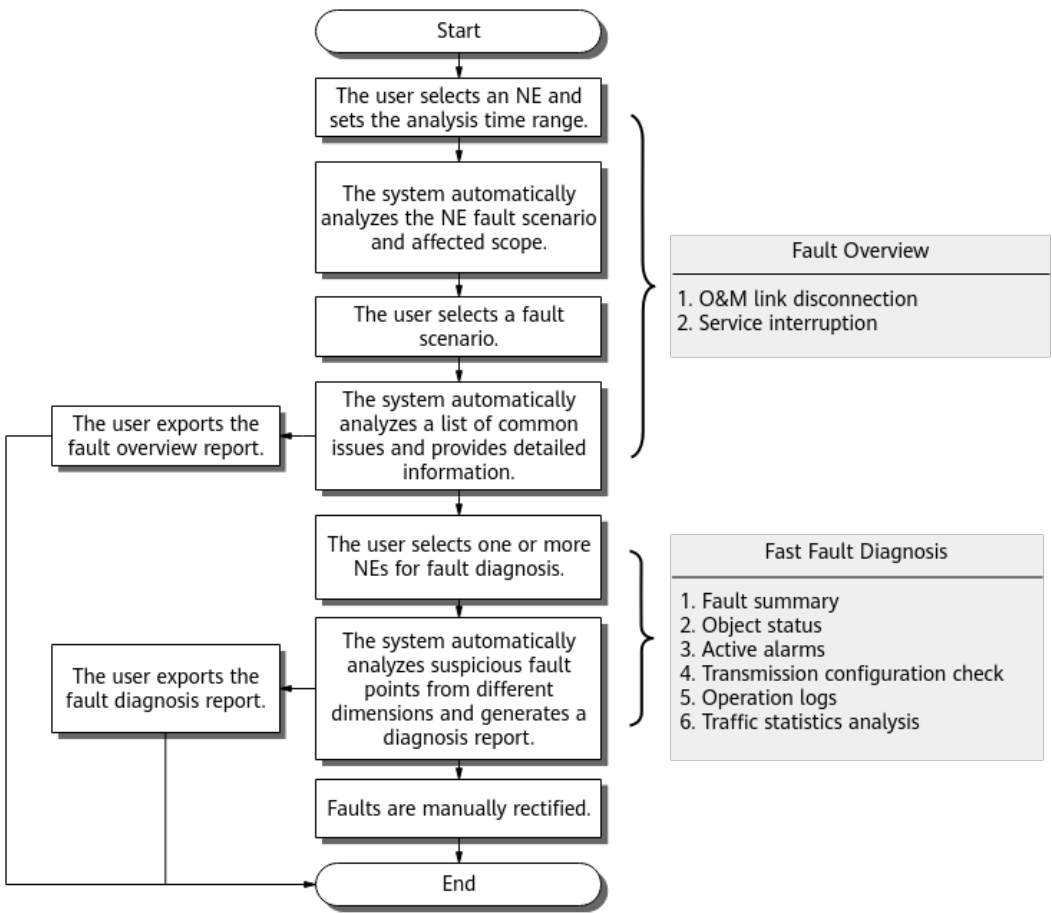
3.1.3 Transmission Fault Management

FMA transmission fault management is used to delimit and locate a fault at the transmission layer of a radio access network (RAN).

- [3.1.3.1 Fault Overview](#)
- [3.1.3.2 Fast Fault Diagnosis](#)
- [3.1.3.3 Assistant Fault Analysis](#)
- [3.1.3.4 Information Collection](#)
- [3.1.3.5 Automatic Fault Detection](#)
- [3.1.3.6 Auxiliary Fault Recovery](#)
- [3.1.3.7 Fault Recovery Confirmation](#)

[Figure 3-31](#) shows the FMA transmission fault diagnosis process.

Figure 3-31 FMA transmission fault diagnosis process



Transmission fault management applies to GSM, UMTS, LTE, and NR. [Table 3-44](#) lists the NE types that support transmission fault management.

Table 3-44 NE types supporting transmission fault management

NE Type	Model
Macro base stations	3900 and 5900 series base stations
LampSite base stations	DBS3900 LampSite and DBS5900 LampSite
Base station controllers	BSC6900 BSC6910

3.1.3.1 Fault Overview

Fault overview provides a transmission network fault status overview. The base station controller fault overview and base station fault overview provide the overall situation of the faulty interfaces for these two NEs, including the interface status, configuration information, NE RAT, and fault occurrence time. Based on the information, users can determine the extent of the damage to the NE transmission interface.

Fault overview provides the following fault scenarios: base station O&M link disconnection, base station service interruption (O&M normal), fault trend monitoring for base station O&M link disconnection or service interruption, base station controller O&M link disconnection, and base station controller service interruption (O&M normal). In each fault scenario, list of common issues, reporting time distribution of active alarms, and detailed fault information are displayed.

A fault overview table provides the distribution of faults on networks (based on the list of common issues) and NEs.

- The detailed information table of base stations applies to GSM, UMTS, LTE, and NR, and provides the fault information about the O&M, S1, Abis, X2 between a gNodeB and an eNodeB, NG between a gNodeB and core network, and Iub interfaces.
- The trend monitoring for base station O&M link disconnection or service interruption is provided.
- The detailed information table of controllers applies to GSM and UMTS and provides the fault information about the O&M, Iu-CS, Iu-PS, A, Gb, and Iur interfaces.

Based on information in [Table 3-45](#), [Table 3-46](#), [Table 3-47](#), [Table 3-48](#), and [Table 3-49](#), the FMA generates a fault overview table listing alarm impacts.

Table 3-45 GSM alarms

Alarm ID	Alarm Name
ALM-21541	SCTP Link Fault
ALM-21551	M3UA Link Fault
ALM-21552	M3UA Destination Entity Route Unavailable
ALM-21553	M3UA Destination Entity Inaccessible
ALM-21512	LAPD Link Fault
ALM-21805	ESL Link Fault
ALM-22003	NSVC Disconnection
ALM-22004	NSVL Faulty
ALM-22005	NSE Faulty
ALM-22009	NSVL Dynamic Configuration Procedure Failure
ALM-21543	SCTP Link IP Address Unreachable
ALM-21821	Site Signaling Link Connection Mismatch
ALM-21581	Path Fault
ALM-21580	IP Path Blocked
ALM-28051	Transmission Link Fault

Alarm ID	Alarm Name
ALM-21605	Path Configuration Error
ALM-21825	CSL Fault
ALM-21807	OML Fault
ALM-40012	NE Is Disconnected NOTE ALM-301 NE Is Disconnected is reported by default for a disconnected NE. If Locate NE Based on Disconnection Alarm is enabled for a type of NE, an NE of this type reports the ALM-40012 NE Is Disconnected alarm when it is disconnected from the OSS. For detailed operations, see General Operations > Alarm Maintenance Tool Management > Locate NE Based on Disconnection Alarm in <i>OSS Maintenance Tool</i>.

Table 3-46 UMTS alarms

Alarm ID	Alarm Name
ALM-21541	SCTP Link Fault
ALM-21543	SCTP Link IP Address Unreachable
ALM-21521	SCCP Subsystem Prohibited
ALM-21551	M3UA Link Fault
ALM-21552	M3UA Destination Entity Route Unavailable
ALM-21553	M3UA Destination Entity Inaccessible
ALM-21581	Path Fault
ALM-21393	Adjacent Node IP Path Ping Failure
ALM-21580	IP Path Blocked
ALM-22214	NodeB Unavailable
ALM-22204	NCP Faulty
ALM-22205	CCP Faulty
ALM-22213	CCP Information Inconsistency Between the RNC and the NodeB

Alarm ID	Alarm Name
ALM-40012	NE Is Disconnected NOTE ALM-301 NE Is Disconnected is reported by default for a disconnected NE. If Locate NE Based on Disconnection Alarm is enabled for a type of NE, an NE of this type reports the ALM-40012 NE Is Disconnected alarm when it is disconnected from the OSS. For detailed operations, see General Operations > Alarm Maintenance Tool Management > Locate NE Based on Disconnection Alarm in <i>OSS Maintenance Tool</i> .

Table 3-47 LTE alarms

Alarm ID	Alarm Name
ALM-25888	SCTP Link Fault
ALM-29201	S1 Interface Fault
ALM-25954	User Plane Fault
ALM-29213	eNodeB S1 Control Plane Transmission Interruption
ALM-25902	Remote Maintenance Link Running Data and Configuration Mismatch
ALM-40012	NE Is Disconnected NOTE ALM-301 NE Is Disconnected is reported by default for a disconnected NE. If Locate NE Based on Disconnection Alarm is enabled for a type of NE, an NE of this type reports the ALM-40012 NE Is Disconnected alarm when it is disconnected from the OSS. For detailed operations, see General Operations > Alarm Maintenance Tool Management > Locate NE Based on Disconnection Alarm in <i>OSS Maintenance Tool</i> .

Table 3-48 NR alarms

Alarm ID	Alarm Name
ALM-25888	SCTP Link Fault
ALM-25954	User Plane Fault
ALM-29800	gNodeB X2 Interface Fault
ALM-29815	gNodeB NG Interface Fault
ALM-29816	gNodeB NG Control Plane Transmission Interruption
ALM-25902	Remote Maintenance Link Running Data and Configuration Mismatch

Alarm ID	Alarm Name
ALM-40012	NE Is Disconnected NOTE ALM-301 NE Is Disconnected is reported by default for a disconnected NE. If Locate NE Based on Disconnection Alarm is enabled for a type of NE, an NE of this type reports the ALM-40012 NE Is Disconnected alarm when it is disconnected from the OSS. For detailed operations, see General Operations > Alarm Maintenance Tool Management > Locate NE Based on Disconnection Alarm in <i>OSS Maintenance Tool</i>.

Table 3-49 NE disconnection alarms on the MAE

Alarm ID	Alarm Name
ALM-301	NE Is Disconnected

Fault overview offers the following functions:

- Supports the export of faults for forwarding and offline review.
- Supports scenario-specific cross-domain information export of faulty NEs (including transmission path information and LLDP notification information report) to facilitate joint troubleshooting of radio access networks and transmission networks. LLDP stands for Link Layer Discovery Protocol. It is a layer 2 discovery protocol defined in IEEE 802.1ab. With the LLDP, the OSS can rapidly obtain the layer 2 network topology information and changes. LLDP must be enabled on both the NE and the FMA to take effect. For details, see [3.4.3.1 Activation](#).
- Supports fast fault diagnosis and fault information collection for selected NEs.
- Allows users to export key transmission data used for node configuration. If transmission faults occur in a large number of sites, transmission services are affected. In this way, key transmission configuration information of related NEs can be quickly generated. This helps wireless and fault handling personnel to demarcate and locate faults.
- Supports fault trend monitoring in base station O&M link disconnection and service interruption (O&M normal) scenarios.
 - Fault trend monitoring shortens the time for determining transmission fault scope from hours to minutes.
 - The fault trend monitoring function displays the total number of NEs, number of faulty NEs beyond the time window, number of disconnected NEs in the time window, and number of recovered NEs.
 - This function can also display the trend of the number of disconnected NEs within a specified time range.
 - The trend of the number of disconnected NEs and the number of recovered NEs can be monitored since the time the trend monitoring task is started.

- On the **Base Station OM Link Disconnection** tab page, the numbers of normal neighboring base stations are displayed in the **X2/Xn Normal Nbr Site Count** column for the base stations disconnected from the OSS respectively. This helps users select the disconnected base stations whose service status can be confirmed for diagnosis. Users can click the sorting button next to **X2/Xn Normal Nbr Site Count** to display the numbers in descending order. It is recommended that only disconnected base stations with the top **X2/Xn Normal Nbr Site Count** values (greater than 0) be selected for diagnosis and analysis.
- Allows transmission configuration data to be classified based on different scenarios: signaling plane fault, user plane fault, and OM fault, as shown in [Table 3-50](#), [Table 3-51](#), and [Table 3-52](#).

Table 3-50 Signaling plane fault scenario

Information Type	Name	Description
S1-C basic information	S1INTERFACE	S1InterfaceId
	CNOPERATOR	CnOperatorId, MCC, and MNC of the S1-C interface
	EPGROUP	EPGROUP ID of the link in the EP model
	SCTPLNK	S1-C link SCTPNO, local and peer IP address, local and peer port numbers (specified by LOCPORT and PEERPORT), and fault information
	IPRT	Next hop information of the S1-C interface
	VLAN and QoS information	VLAN ID, VLANPRIO, and DSCP value of the S1-C interface
IPsec-related information displayed in the IPsec scenario	IPsec Flag	YES/NO: Indicates whether the link is encrypted by IPsec.
	DEVIP	If the IPsec networking is used, the physical IP address of the link (DEVIP, when PT is ETH) is displayed.
	IKEPEER	SeGW IP address and IKEVERSION
	IKEPROPOSAL	PROPID and AUTHMETH (PSK/IKE displayed)
	ACLRULE	ACLID and RULEID specify which ACL rule to be matched.
	APPCERT	Name and expiration date of the certificate used by the current IPsec tunnel

Information Type	Name	Description
Physical port information	ETHPORT	Port number, port attribute (optical/electrical), MAC address, and MTU

Table 3-51 Signaling plane fault scenario

Information Type	Name	Description
S1-U basic information	CNOPERATOR	CnOperatorId, MCC, and MNC of the S1-U interface
	EPGROUP	EPGROUP ID of the link in the EP model
	USERPLANEHOST & USERPLANEPEER or IPPATH	Local/Peer IP address of the S1-U link and fault information (obtained by running the DSP EPGROUP command)
	GTP-U static detection switch status	Whether STATICCHK is enabled for GTP-U static detection
	IPRT	Next hop information of the S1-C interface
	VLAN and QoS information	VLAN ID, VLANPRIO, and DSCP value of the S1-U interface
IPsec-related information displayed in the IPsec scenario	IPsec Flag	YES/NO: Indicates whether the link is encrypted by IPsec.
	DEVIP	If the IPsec networking is used, the physical IP address of the link (DEVIP, when PT is ETH) is displayed.
	IKEPEER	SeGW IP address and IKEVERSION
	IKEPROPOSAL	PROPID and AUTHMETH (PSK/IKE displayed)
	ACLRULE	ACLID and RULEID specify which ACL rule to be matched.
	APPCERT	Name and expiration date of the certificate used by the current IPsec tunnel
Physical port information	ETHPORT	Port number, port attribute (optical/electrical), MAC address, and MTU

Table 3-52 O&M fault scenario

Information Type	Name	Description
OMCH basic information	OMCH configuration information	Active/standby flag, local and peer IP addresses, and whether routes are bound
	LST EMS	The IP address of the connected MAE is displayed.
	IPRT	Next hop information of the OMCH
	VLAN and QoS information	VLAN ID, VLANPRIO, and DSCP value of the OMCH
	SSL-related information	Indicates whether the SSL connection is used and which authentication mode is used.
	APPCERT	Name and expiration date of the certificate used by the SSL connection
IPsec-related information displayed in the IPsec scenario	IPsec Flag	YES/NO: Indicates whether the link is encrypted by IPsec.
	DEVIP	If the IPsec networking is used, the physical IP address of the link (DEVIP, when PT is ETH) is displayed.
	IKEPEER	SeGW IP address and IKEVERSION
	IKEPROPOSAL	PROPID and AUTHMETH (PSK/IKE displayed)
	ACLRULE	ACLID and RULEID specify which ACL rule to be matched.
	APPCERT	Name and expiration date of the certificate used by the current IPsec tunnel
Physical port information	ETHPORT	Port number, port attribute (optical/electrical), MAC address, and MTU

3.1.3.2 Fast Fault Diagnosis

Users can find faulty NEs based on the fault overview of FMA transmission fault management, and then start fast fault diagnosis (also called one click test) for transmission faults that persist on a network. Faults are diagnosed through segment-based pings and trace routes, radio configuration check, and status query. Based on the diagnosis, an analysis report is generated to help users rectify the faults, improving fault recovery efficiency. The fault diagnosis covers port faults, IP address segment faults, and cross-NE or single-NE transmission configuration check exceptions.

Fast fault diagnosis includes the following tab pages: **One Click Test Tasks**, **Intelligent Continuity Check(User/Signaling Plane)**, **Intelligent Continuity Check(OM/Clock Plane)**, **Custom Continuity Check (OSS)**, and **Custom Trace Route(OSS)**.

Fault Diagnosis

Users can view the one-click diagnosis report on the **One Click Test Tasks** tab page. One-click diagnosis report provides comprehensive analysis of the on-site fault information (see [Table 3-53](#)).

Table 3-53 One-click fault diagnosis report

Parameter	Description
Summary	Displays all topologies (lub, Abis, S1, gNodeB X2, and O&M interfaces) of the faulty NE (base station) or some topologies (lur, lu-PS, lu-CS, A, Gb, and O&M interfaces) of the faulty NE (base station controller). Each functional module reports the suspicious fault scope or fault point based on the analysis and detection results. If there is no fault, no alarm is reported. The summary part displays the conclusion content and the reported module name (such as segment-based ping, radio configuration check, and status query), helping users view details.
Segment-based pings and trace routes	Displays segment-based ping results by interfaces (lu-PS, lu-CS, lur, A, or Gb) and planes (control plane, user plane, or O&M plane). If the ping operation fails, the trace route operation is performed to further demarcate the fault. For segment-based pings and trace routes, a transmission pool contains multiple pairs of interface boards and multiple pairs of IP addresses. The FMA performs ping and trace route operations on IP addresses in use. Currently, the transmission pool mode applies to the lub interface. In multi-operator sharing scenarios, faulty links of various operators can be detected simultaneously through segment-based pings and trace routes. Only one faulty link is detected for each operator, and multiple faulty links of the same base station are displayed in different lines.

Parameter	Description
Active alarm	Displays transmission alarms by NEs to help users quickly locate the root cause of the fault. Displays all alarms related to a fault on the MAE: • Base station alarms for 4G and 5G NEs • 2G base station alarms and alarms related to the Abis interface for base station controllers • 3G base station alarms and alarms related to the Iub interface for base station controllers • Alarms related to the Iu-CS, Iu-PS, Iur, A, and Gb interfaces for base station controllers • NE disconnection alarms
Object status	• For a base station connected to the OSS: Displays the status information about the transmission objects on the base station. The information includes the configurations, available status, IP addresses, and subnet masks of the objects. • For a base station disconnected from the OSS: Displays the service load information about the base station, including its hardware load, transmission load, and air interface load. Such information is obtained by delivering MML commands to neighboring base stations that have X2/Xn links to the disconnected base station as well as normal O&M channels.
Radio configuration check	Automatically checks exceptions in cross-NE or single-NE transmission configuration, including but not limited to bandwidth configuration consistency and VLAN priority.
Operation log	Records MAE-Access operation logs and MML logs generated eight hours before and after a fault occurs during the one-click fault diagnosis, as well as the MAE-Deployment logs generated over the past two hours on the server.
Traffic statistics	Displays counter values of base stations or base station controllers. Transmission counters are included in an overall tab page. One tab page is provided for each function subset, displaying counter values of all measurement periods for the current NE within 24 hours.
Fault tree	Provides fault tree branches for auxiliary fault recovery. The fault scenarios include O&M link disconnection and service interruption.

Parameter	Description
OMCH Maintenance Info	About 10 to 30 minutes after the base station is disconnected, it reports the maintenance and test information of the disconnected OMCH to the OSS through DHCP. The information is displayed by NE for O&M personnel to analyze the cause of the disconnection and the running status and behavior of the base station in order to take proper recovery measures in a timely manner. The following information is included: • DHCP detection information • OMCH parameter settings • OMCH status information • OMCH self-recovery information The handling suggestions are as follows: • If configurations are modified before OMCH disconnection, check whether the disconnection is caused by the configuration modification. • If configurations are not changed, check the link status. If the link status is normal, check the number of sent and received packets to determine the cause of the disconnection. • Check the self-healing measures that have been taken and the number of self-healing attempts. If the number exceeds three, local onsite maintenance may be required. If no information is displayed in the OMCH Maintenance Info area in the diagnosis result of the transmission FMA, the possible causes are as follows: • The DHCP port of the OSS is not enabled. • No DHCP packet is received from the NE. • Other issues occur. Contact Huawei technical support.

This function can diagnose faults in a maximum of 25 sites at a time. Users can export the one-click fault diagnosis report in the html format for offline view.

NOTE

- After report generation, ten reports can be exported within 10s.
- The X2 interface information is not displayed in the one-click fault diagnosis report due to the large quantity of X2 interfaces.

Connectivity Check

Connectivity check is used by O&M personnel when fault diagnosis cannot automatically diagnose system faults in one-click mode. This function also helps O&M personnel quickly check the overall system transmission status when no fault exists in the system. Connectivity check includes network quality detection and provides the following methods to help O&M personnel quickly demarcate transmission problems.

Table 3-54 Description of the ping test

Function	Description	Impact on Services	Operation Specifications
Intelligent Continuity Check(User/ Signaling Plane)	Checks the connectivity of IP links between base station controllers and peer NEs (such as GBTSS, NodeBs, base station controllers, and core network NEs), between eNodeBs and peer NEs (such as core network NEs), between gNodeBs and peer NEs (such as eNodeBs and core network NEs), and between USNs and peer NEs (such as eNodeBs and gNodeBs). Base station controllers, eNodeBs, gNodeBs, or USNs are at the source end, and peer NEs are at the destination end. This function allows you to: • Select a base station controller, a USN, or multiple eNodeBs or gNodeBs at the source end and select multiple NEs at the destination end to start ping tests. NOTE You can select a maximum of 2000 eNodeBs or gNodeBs at a time at the source end to start ping tests in batches. • Set the ping direction. – If destination-end NEs are base station controllers, core network NEs, GBTSS, eNodeBs, gNodeBs, or virtual NEs, you can set the ping direction to Source to Destination. – If destination-end NEs are NodeBs, you can set the ping direction to Source to Destination, Destination to Source, or Bidirectional.	No	Each NE supports only one check command at a time.
Intelligent Continuity Check(OM/ Clock Plane)	Checks the connectivity of IP links between a base station (such as a NodeB, an eNodeB, or a gNodeB) and a peer NE (MAE). The main functions include setting the ping packet quantity, ping detection duration, ping packet size, timeout value, DSCP, packet sending interval, and DF switch.		

Function	Description	Impact on Services	Operation Specifications
Custom Continuity Check(OSS)	Checks the connectivity between the IP link from the OSS IP address to a specified IP address or the IP address of the O&M channel for a specified NE. The main functions are as follows: • The source end is the OSS IP address by default. The peer end can be a specified IP address or the IP address of the O&M channel of a specified NE. NOTE Multiple NEs (a maximum of 400 NEs) can be selected at a time at the peer end. • OSS initiates a ping test to the specified IP address and displays the results. The ping packet quantity is 3. The ping packet size, timeout value, and packet sending interval use default values.		
Custom Trace Route(OSS)	Detects and locates connectivity faults by segment. The main functions are as follows: • The trace route operation is initiated from the master server by default. • The source end is the OSS IP address by default. • The peer end can be a specified IP address or the IP address of the O&M channel of a specified NE. NOTE • Multiple NEs (a maximum of 400 NEs) can be selected at a time at the peer end. • The trace route connectivity check supports a maximum of six hops.		

3.1.3.3 Assistant Fault Analysis

An assistant fault analysis function applies when the fast fault diagnosis function cannot automatically diagnose system faults in one-click mode. It also helps users quickly check the overall system transmission status when no fault occurs on the transmission network. This function consists of 5-dimensional topology display and network detection.

3.1.3.4 Information Collection

Onsite information needs to be collected for troubleshooting. This function can quickly and accurately collect onsite information and improve fault recovery efficiency. Information collection for transmission faults only applies to eNodeBs and gNodeBs, as shown in [Table 3-55](#).

Table 3-55 Information collection for transmission faults

Collection Item	Scope
NE configuration file	One file for each NE, five NEs in total
MML query output	One file for each NE, five NEs in total
Results of pinging a base station from the MAE	One file for each NE, five NEs in total
Results of tracing routes from the MAE to a base station	One file for each NE, five NEs in total
KPI data	During a specified collection period
Alarm data	During a specified collection period
MAE operation logs	During a specified collection period
Original operation logs of top 3 NEs	One file for each NE, three NEs in total

3.1.3.5 Automatic Fault Detection

The FMA can automatically detect transmission faults and performs fault diagnosis and information collection after detection:

- The FMA selects top faulty sites, performs fault diagnosis, and generates a diagnosis report. The diagnosis reports triggered by automatic detections and manually triggered are consistent, expect that the former report does not include topology information. For details, see [3.1.3.2 Fast Fault Diagnosis](#).
- The data collected in fault information collections triggered by automatic detections and that manually collected on the FMA is consistent. For details, see [3.1.3.4 Information Collection](#).

Users can log in to the FMA to view and download diagnosis reports and transmission fault information.

3.1.3.6 Auxiliary Fault Recovery

Auxiliary fault recovery provides fault recovery assistance for specific configuration faults displayed in the fast fault diagnosis results. Auxiliary fault recovery provides the following functions:

- Displaying the historical configuration change track of the faulty NE: After the fast fault diagnosis function identifies a faulty NE, the FMA displays all the configuration changes made to the NE two hours before the fault occurs. This helps users determine whether the fault is caused by changes in configurations.
- Editing of auxiliary fault recovery scripts: The FMA provides a window for editing auxiliary fault recovery scripts. Users can edit the MOCs and parameters to be delivered to faulty NEs on the GUI. In addition, users can select a configuration change record from the historical configuration change track, and reconfigure specific parameters.

- Generating auxiliary fault recovery scripts: The FMA can interconnect with the MAE-Deployment to generate and display auxiliary recovery scripts based on the MOCs and parameters that users specify in the script editing window.

Table 3-56 lists fault tree branches supported by auxiliary fault recovery.

Table 3-56 Fault tree branches supported by auxiliary fault recovery

Scenario	Fault Tree Branch
OM Disconnection	None
NE Service Interruption	NE Service Interruption > NE Report IKE Negotiate Fail Alarm

3.1.3.7 Fault Recovery Confirmation

The FMA supports fault recovery confirmation in fast fault recovery scenarios. It showcases the changes in the status of transmission ports and transmission alarms using visualized graphics on the dashboard after the recovery plan is executed. Fault handling personnel can use such data to determine whether a fault is rectified. **Table 3-57** lists the detailed functions of fault recovery confirmation.

Table 3-57 Functions of fault recovery confirmation

Function	Sub-Function	Description
Fault recovery confirmation	List of transmission alarm status	Displays the alarm clearance status after the recovery plan is executed, helping users to determine whether the fault is resolved.
	List of transmission port status	Displays the status of the transmission port instance after the recovery plan is executed, helping users to determine whether the fault is recovered.
	Saving of the recovery confirmation results	Saves the transmission fault alarm clearance results and transmission ports status result.

OM Emergency Channel-based Self-Healing

When the OMCH of the base station remains interrupted for a period of time, the base station reports the SCTP link status to the MAE through the OM emergency channel. Users can determine whether to enable OM emergency channel-based self-healing based on the SCTP link status. If users select **Enable**, the MAE enables OM emergency channel-based self-healing for 2 hours and then disables the channel automatically. During this period, if the base station sends an OM channel

self-healing request again, the MAE responds with an acknowledgment message. The base station forcibly resets after receiving the acknowledgment message from the MAE. If the OM channel recovers after the base station is reset, the OM emergency channel-based self-healing is complete. Otherwise, the base station enters the minimum configuration mode and starts the automatic OM channel establishment procedure. For details, see *Automatic OMCH Establishment*.

To enable the OM emergency channel-based self-healing function, perform the following operations on the base station and MAE sides:

- Step 1** On the base station side: Select the **OMCH_ESCAPE_SW** option of the **TRANSALGEXTPARA.OMCHMNGSW** parameter. This option is selected by default. To disable this function, run the **SET TRANSALGEXTPARA** command with the **OMCH_ESCAPE_SW** option of the **TRANSALGEXTPARA.OMCHMNGSW** parameter deselected. After the OM emergency channel is disabled on the base station, OM emergency channel-based self-healing cannot be enabled through the MAE.
- Step 2** On the MAE side: Log in to the MAE-Access, choose **FMA > Transmission > Fault Recovery Confirmation > OM Emergency Channel (OM Link Disconnection)**, and click **Query**. Select the base station for which OM emergency channel-based self-healing needs to be enabled and click **Enable**.

----End

This function is not supported when the base station software running mode is set to **EMERGENCYMODE**.

⚠ CAUTION

- If the OM channel of the base station is interrupted for more than 40 minutes, the base station reports the SCTP link status to the MAE through the OM emergency channel. If the SCTP link status cannot be queried on the MAE, you are advised to perform troubleshooting on the intermediate transport network or contact Huawei technical support.
 - Enabling the OM emergency channel (OM link disconnection) is a high-risk operation, which will cause the base station to reset or enter the minimum configuration mode. Perform this operation under the guidance of Huawei engineers.
-

3.2 Network Analysis

3.2.1 Benefits

- Facilitates fault handling and shortens the duration of the fault impact on network services, ensuring stable network operation.
- Improves the network or NE fault location efficiency and reduces O&M costs.
- Lowers skill requirements for fault handling.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

This function has no impact on the network.

3.3 Requirements

3.3.1 Licenses

N/A

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

None

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

None

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

The BTS5929R does not support this feature.

[Table 3-58](#) lists the MAE hardware and networking scenarios that support this feature.

Table 3-58 MAE hardware and networking scenarios that support FMA feature

Platform	Server Type	Scenario	Supported in This Scenario	Deployment Condition
Virtual	N/A	Cluster System	Yes	This function is enabled by default.

3.3.4 Others

Account and Permission

An FMA user account must be created. For details, see [3.4.1 When to Use](#).

Port Requirements

This feature requires that related ports be enabled on the MAE. For details, see *MAE Communication Matrix* in *iMaster MAE Product Documentation*.

3.4 Operation and Maintenance

3.4.1 When to Use

It is recommended that the FMA be enabled in the post-upgrade health check, accident drills, and accident handling on the live network.

The hardware required by the FMA feature is automatically deployed together with the MAE. If the number of equivalent NEs does not exceed 1200, the FMA can be deployed together with the MAE. If the number of equivalent NEs is greater than 1200, an independent board is required to deploy the FMA.

The FMA is deployed through software and the deployment process is as follows:

- Create the FMA user account on the MAE and grant user permissions. For details, see *MAE Function Description About FMA* in *iMaster MAE Product Documentation*.
To switch to the Web LMT for GSM and UMTS NEs, you need to add device types to the management domains of the FMA user on the MAE client as required, such as **BSC6900GSM**, **BSC6900GU**, **BSC6900UMTS**, **BSC6910GSM**, **BSC6910GU**, and **BSC6910UMTS**.

NOTE

The FMA software can be separately upgraded to a later R/C version for GSM and UMTS NEs to accelerate urgent fault recovery on the live network.

The FMA and MBSC must be of the same R/C version. For example, if the MBSC version is V900R018C10SPC530, the FMA version must be V900R018C10 FMA Rules Update Packet.

The O&M personnel can obtain the separate software package to upgrade the FMA from support.huawei.com. Choose **Software > Wireless Network > Wireless Network Common > SingleRAN > SingleRAN_O&M tools**. Click the hyperlink of the required FMA software package and download it using a Huawei engineer account.

The FMA software can be installed and uninstalled only by running MML commands.

- For details on how to start the FMA web client on the MAE, see "Logging In to the FMA System" in *iMaster MAE Product Documentation*.
- For details on how to initialize the FMA (configure the data synchronization information between the FMA and the MAE), see "How Do I Set Data Synchronization Between FMA and MAE?" in *iMaster MAE Product Documentation*.

For deployment process details, see "Deployment Process" in *iMaster MAE Product Documentation*.

3.4.2 Precautions

- If the available resources on the MAE cannot meet the minimum operating requirements, the feature cannot be enabled.
- If a large number of MAE resources are consumed, the FMA feature is forcibly disabled to prevent interruption of the MAE operation. If the feature is disabled during fault handling, the troubleshooting process is forcibly stopped. After resources are released, restart the FMA service. For details, see "FMA Commands" in *MAE Command Reference*.
- The FMA feature is available only when data processing and calculation are complete using a data configuration account. A maximum of 48 hours are required for data processing and calculation, depending on network data volume.

3.4.3 Data Configuration

This section describes the deployment of Fault Management Assistant (FMA) on the MAE.

In an integrated deployment, GSM and UMTS wireless fault management is enabled on the LMT by default. For details, see *BSC6900 GU LMT User Guide* and *BSC6910 GU LMT User Guide*.

3.4.3.1 Activation

The FMA feature is available after the MAE is deployed.

Perform the following operations on both the FMA and NE to activate the LLDP function:

On the FMA:

Step 1 Click **Synchro Settings**.

Step 2 Select **Subscribe LLDP Info**.

----End

On the base station controller and base station:

- Using the MAE-Deployment
On the MAE-Deployment, set **LLDP Notify Switch** in the **LLDPGLOBAL** MO to **ENABLE**.
 - Set port parameters in the **LLDPLOCAL** MO when the old transmission configuration model is used.
 - Set port parameters in the **LLDP** MO when the new transmission configuration model is used.For detailed operations on the MAE-Deployment, see "MAE-Deployment Operation and Maintenance" in *iMaster MAE Product Documentation*.
- Using MML commands
 - a. Run the **SET LLDPGLOBALINFO** command. Set **NOTIFYSW (LTE eNodeB, 5G gNodeB)** to **ENABLE** and specify other LLDP global parameters.
 - b. Run the following command to add a port through which LLDP messages are reported.
 - **ADD LLDP** when the old transmission configuration model is used
 - **ADD LLDP** when the new transmission configuration model is used

3.4.3.2 Deactivation

The FMA feature is deactivated on the MAE OSS self-maintenance unit (OSMU). For details, see "FMA Commands" of *MAE Command Reference* in *iMaster MAE Product Documentation*.

Perform the following operations on both the NE and the FMA to deactivate the LLDP function:

On the base station controller and base station:

Step 1 Run the **SET LLDPGLOBALINFO** command and set **NOTIFYSW** to **DISABLE(DISABLE)**.

Step 2 Run the following commands to delete the information about the LLDP port.

- **RMV LLDP** when the old transmission configuration model is used
- **RMV LLDP** when the new transmission configuration model is used

----End

After logging in to the FMA system, perform the following operations:

Step 1 Click **Synchro Settings**.

Step 2 Deselect **Subscribe LLDP Messages**.

----End

3.4.3.3 Using the MAE-Deployment

N/A

3.4.4 Activation Verification

After the FMA feature is enabled, you can use the following methods to verify if the FMA is operating properly:

- Check whether the FMA is operating properly on the MAE-OSMU. For details, see **Checking the Status of MAE and Database Services** in *iMaster MAE Product Documentation*.
- Check whether you can switch from the MAE client to the FMA web client. For details, see **Logging In to the FMA System** in *iMaster MAE Product Documentation*.
- Note that you must wait for up to 48 hours to use the FMA feature after the data is synchronized between the FMA and the MAE. The duration varies depending on the number of NEs managed by the MAE. For details, see **How Do I Set Data Synchronization Between FMA and MAE?** in *iMaster MAE Product Documentation*.
- Check if you can start fault diagnosis on the wireless fault management module.
- Check if you can start one-click fault diagnosis on the transmission fault management module.
- Check if you can view basic information about peer devices in the network topology. If this function is supported, the LLDP function takes effect.

3.4.5 Network Monitoring

When the FMA feature is in operation, users can check whether the FMA service is performing properly on the MAE-OSMU.

When a fault occurs on the FMA, no alarm is generated on the MAE client. Users can check whether the FMA process and services are performing properly on the MAE-OSMU to determine whether the FMA is faulty. For details, see "Checking the Status of MAE and Database Services" in *iMaster MAE Product Documentation*.

4 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary* in *iMaster MAE Product Documentation*.

5 Reference Documents

- *iMaster MAE Product Documentation*
- *MAE Command Reference*
- *OSS Maintenance Tool*
- *MAE-Deployment Operation and Maintenance*
- *SRAN Networking and Evolution Overview*
- *Automatic OMCH Establishment*
- *Synchronization in eRAN Feature Documentation*
- *Synchronization in 5G RAN Feature Documentation*
- *BSC6900 GU LMT User Guide*
- *BSC6910 GU LMT User Guide*